

closed cams are not ground after heat treatment and are used as-milled. Though the conjugate cam approach avoids the groove tolerance and heat treat distortion problems, there are still two matched cam surfaces to be made per cam. Thus, the desmodromic cam's dynamic advantages come at a significant cost premium.

We will now repeat the cam design of Example 15-1, modified for desmodromic operation. This is simple to do with program DYNACAM. We will merely specify the spring constant and preload values to be zero, which then assumes that the follower train is a rigid body. A more accurate result can be obtained by calculating and using the effective spring constant of the combination of parts in the follower train, once their geometries and materials are defined. The dynamic forces will now be negative as well as positive, but a form-closed cam can both push and pull.

EXAMPLE 15-2

Dynamic Force Analysis of a Form-Closed (Desmodromic) Cam-Follower System.

Given: A translating roller follower as shown in Figure 15-11a is driven by a form-closed radial plate cam which has the following program:

Segment 1: Rise 1 inch in 50° with modified sine acceleration
 Segment 2: Dwell for 40°
 Segment 3: Fall 1 inch in 50° with cycloidal displacement
 Segment 4: Dwell for 40°
 Segment 5: Rise 1 inch in 50° with 3-4-5 polynomial displacement
 Segment 6: Dwell for 40°
 Segment 7: Fall 1 inch in 50° with 4-5-6-7 polynomial displacement
 Segment 8: Dwell for 40°
 Camshaft angular velocity is 18.85 rad/sec.
 Follower effective mass is 0.0738 in-lb-sec² (blobs).
 Damping is 15% of critical ($\zeta = 0.15$).

Problem: Compute the dynamic force function for the cam. Keep the pressure angle under 30° .

Solution:

- 1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for the specified cam functions. The acceleration for this cam is shown in Figure 15-7 (p. 701) and has a maximum value of 3504 in/sec². See Chapter 8 to review this procedure.
- 2 Calculate radius of curvature and pressure angle for trial values of prime circle radius, and size the cam to control these values. Figure 15-8 shows the pressure angle function and Figure 15-9 the radii of curvature for this cam with a prime circle radius of 4 in and zero eccentricity. The maximum pressure angle is 29.2° and the minimum radius of curvature is 1.7 in. Figure 8-47 (p. 407) shows the finished cam profile. See Chapter 8 to review these calculations.
- 3 With the kinematics of the cam defined, we can address its dynamics. To solve equation 15.10 (p. 702) for the cam force, we assume zero values for the spring constant k and the preload F_{pl} . The value of c is assumed to be the same as in the previous example, 1.153. The kinematic parameters are known.

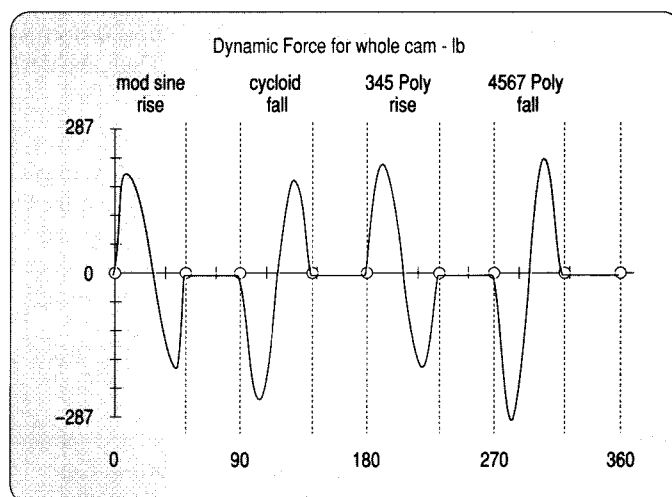


FIGURE 15-12

Dynamic force in a form-closed cam-follower system

- 4 Program DYNACAM does this computation for you. The dynamic force that results is shown in Figure 15-12. Note that the force is now more nearly symmetric about the axis and its peak absolute value is 289 lb. Crossover shock will occur each time the follower force changes sign. Open the file E15-02.cam in DYNACAM to see this example.

Compare the dynamic force plots for the force-closed system (Figure 15-10b, p. 703) and the form-closed system (Figure 15-12). The absolute peak force magnitude on either side of the track in the form-closed cam is less than that on the spring-loaded one. This shows the penalty that the spring imposes on the system in order to keep the joint closed. Thus, either side of the cam groove will experience lower stresses than will the open cam, except for the areas of crossover shock mentioned on p. 704.

15.5 CAMSHAFT TORQUE

The kinetostatic analysis assumes that the camshaft will operate at some constant speed ω . As we saw in the case of the fourbar linkage in Chapter 11 and with the slider-crank mechanism in Chapter 13, the input torque must vary over the cycle if the shaft velocity is to remain constant. The torque can be easily calculated from the power relationship, ignoring losses.

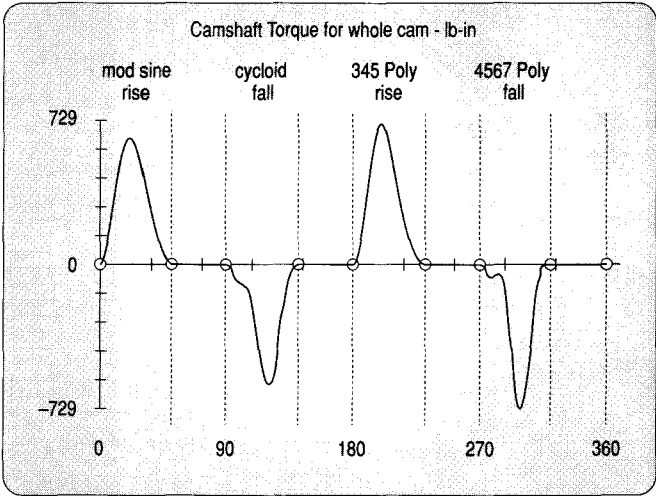
$$\text{Power in} = \text{Power out}$$

$$T_c \omega = F_c v \tag{15.11}$$

$$T_c = \frac{F_c v}{\omega}$$

Once the cam force has been calculated from either equation 15.9 or 15.10, the camshaft torque T_c is easily found since the follower velocity v and camshaft ω are both known. Figure 15-13a shows the camshaft input torque needed to drive the force-closed cam designed in Example 15-1 (p. 700). Figure 15-13b shows the camshaft input torque needed to drive the form-closed cam designed in Example 15-2 (p. 705). Note that the torque required to drive the force-closed (spring-loaded) system is significantly higher than that needed to drive the form-closed (track) cam. The spring force is also extracting a penalty here as energy must be stored in the spring during the rise portions which will tend to slow the camshaft. This stored energy is then returned to the camshaft during the fall portions, tending to speed it up. The spring loading causes larger oscillations in the torque.

(a) Force-closed
(spring-loaded)
cam-follower



(b) Form-closed
(desmodromic)
cam-follower

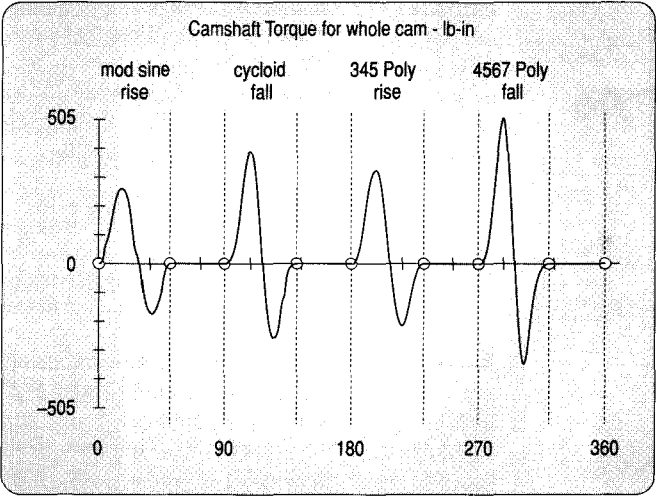


FIGURE 15-13
Input torque in force- and form-closed cam-follower systems

A flywheel can be sized and fitted to the camshaft to smooth these variations in torque just as was done for the fourbar linkage in Section 11.11 (p. 548). See that section for the design procedure. Program DYNACAM integrates the camshaft torque function pulse by pulse and prints those areas to the screen. These energy data can be used to calculate the required flywheel size for any selected coefficient of fluctuation.

One useful way to compare alternate cam designs is to look at the torque function as well as at the dynamic force. A smaller torque variation will require a smaller motor and/or flywheel and will run more smoothly. Three different designs for a single-dwell cam were explored in Chapter 8. (See Examples 8-5, p. 375, 8-6, p. 377, and 8-8, p. 383.) All had the same lift and duration but used different cam functions. One was a double harmonic, one cycloidal, and one a sixth-degree polynomial. On the basis of their kinematic results, principally acceleration magnitude, we found that the polynomial design was superior. We will now revisit this cam as an example and compare its dynamic force and torque among the same three programs.

EXAMPLE 15-3

Comparison of Dynamic Torques and Forces Among Three Alternate Designs of the Same Cam.

Given: A translating roller follower as shown in Figure 15-1 (p. 687) is driven by a force-closed radial plate cam which has the following program:

Design 1

Segment 1: Rise 1 inch in 90° double harmonic displacement

Segment 2: Fall 1 inch in 90° double harmonic displacement

Segment 3: Dwell for 180°

Design 2

Segment 1: Rise 1 inch in 90° cycloidal displacement

Segment 2: Fall 1 inch in 90° cycloidal displacement

Segment 3: Dwell for 180°

Design 3

Segment 1: Rise 1 inch in 90° and fall 1 inch in 90° with polynomial displacement. (A single polynomial can create both rise and fall.)

Segment 2: Dwell for 180°

Camshaft angular velocity is 15 rad/sec. Follower effective mass is 0.0738 in-lb-sec² (blobs). Damping is 15% of critical ($I_c = 0.15$).

Find: The dynamic force and torque functions for the cam. Compare their peak magnitudes for the same prime circle radius.

Solution: Note that these are the same kinematic cam designs as are shown in Figures 8-24 (p. 377), 8-23 (p. 376), and 8-29 (p. 385).

- 1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for each of the specified cam designs. See Chapter 8 to review this procedure.
- 2 Calculate the radius of curvature and pressure angle for trial values of prime circle radius, and size the cam to control these values. A prime circle radius of 3 in gives acceptable pressure angles and radii of curvature. See Chapter 8 to review these calculations.

- 3 With the kinematics of the cam defined, we can address its dynamics. To solve equation 15.1a (p. 687) for the cam force, we will assume a value of 50 lb/in for the spring constant k and adjust the preload F_{pl} for each design to obtain a minimum dynamic force of about 10 lb. For design 1 this requires a spring preload of 28 lb; for design 2, 15 lb; and for design 3, 10 lb.
- 4 The value of damping c is calculated from equation 15.2i (p. 690). The kinematic parameters x , v , and a are known from the prior analysis.
- 5 Program DYNACAM will do these computations for you. The dynamic forces that result from each design are shown in Figure 15-14 and the torques in Figure 15-15. Note that the force is largest for design 1 at 82 lb peak and least for design 3 at 53 lb peak. The same ranking holds for the torques which range from 96 lb-in for design 1 to 52 lb-in for design 3. These represent reductions of 35% and 46% in the dynamic loading due to a change in the kinematic design. Not surprisingly, the sixth-degree polynomial design which had the lowest acceleration also has the lowest forces and torques and is the clear winner. Open the files E08-05.cam, E08-06.cam, and E08-08.cam in program DYNACAM to see these results.

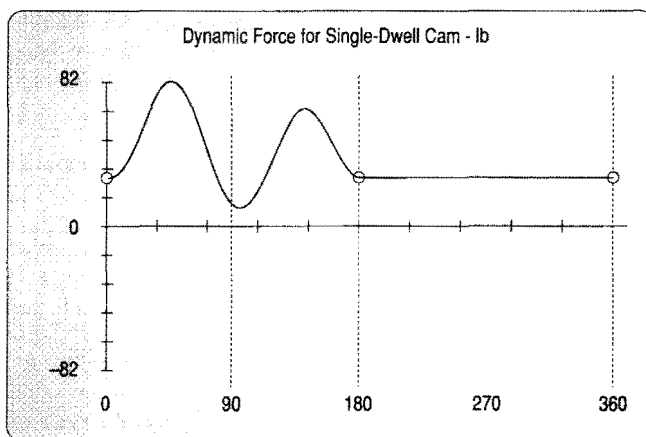
15.6 MEASURING DYNAMIC FORCES AND ACCELERATIONS

As described in previous sections, cam-follower systems tend to be underdamped. This allows significant oscillations and vibrations to occur in the follower train. Section 8.8 (p. 412) discussed the effects of manufacturing errors on the actual accelerations of cam-followers and showed dynamic measurements of acceleration. Dynamic forces and accelerations can be measured fairly easily in operating machinery. Compact, piezoelectric force and acceleration transducers are available that have frequency response ranges in the high thousands of hertz. Strain gages provide strain measurements that are proportional to force and have bandwidths of a kilohertz or better.

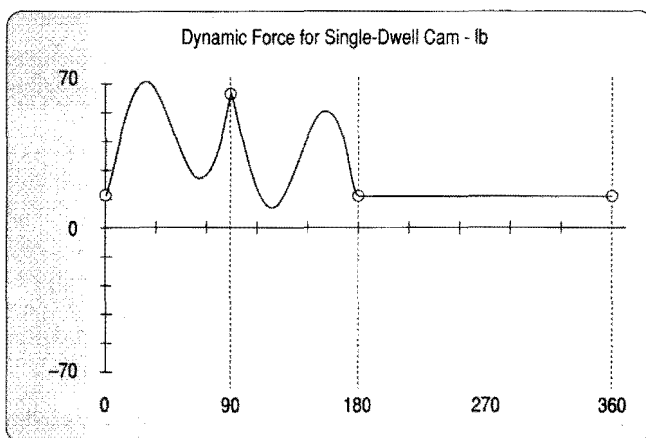
Figure 15-16 shows acceleration and force curves as measured on the follower train of a single overhead camshaft (SORC) valve train in a 1.8-liter four-cylinder inline engine. [2] The nonfiring engine was driven by an electric motor on a dynamometer. The camshaft is turning at 500, 2000, and 3000 rpm (1000, 4000, and 6000 crankshaft rpm), respectively, in the three plots of Figure 15-16a, b, and c. Acceleration was measured with a piezoelectric accelerometer attached to the head of one intake valve, and the force was calculated from the output of strain gages placed on the rocker arm for that intake valve. The theoretical follower acceleration curve (as designed) is superposed on the measured acceleration curve. All acceleration measurements are converted to units of mm/deg^2 (i.e., normalized to camshaft speed) to allow comparison with one another and with the theoretical acceleration curve.

At 500 camshaft rpm, the measured acceleration closely matches the theoretical acceleration curve with some minor oscillations due to spring vibration. At 2000 camshaft rpm, significant oscillation in the measured acceleration is seen during the first positive and in the negative acceleration phase. This is due to the valve spring vibrating at its natural frequency in response to excitation by the cam. This is called "spring surge" and is a significant factor in valve spring fatigue failure. At 3000 camshaft rpm, the spring surge is still present but is less prominent as a percentage of total acceleration. The frequency content of the cam's forcing function passed through the first natural fre-

(a) Double harmonic rise -
double harmonic fall



(b) Cycloidal rise -
cycloidal fall



(c) Sixth-degree
polynomial

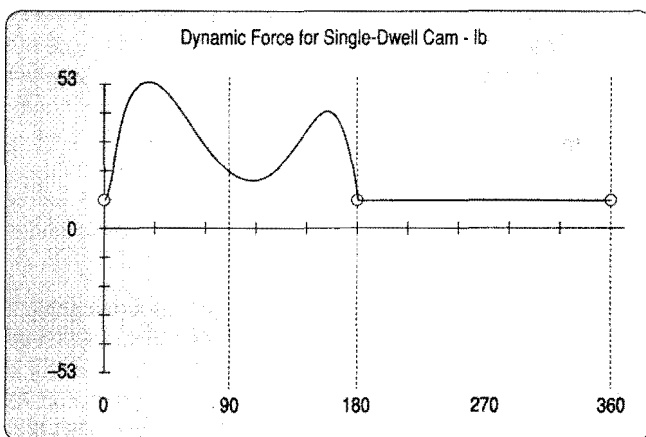
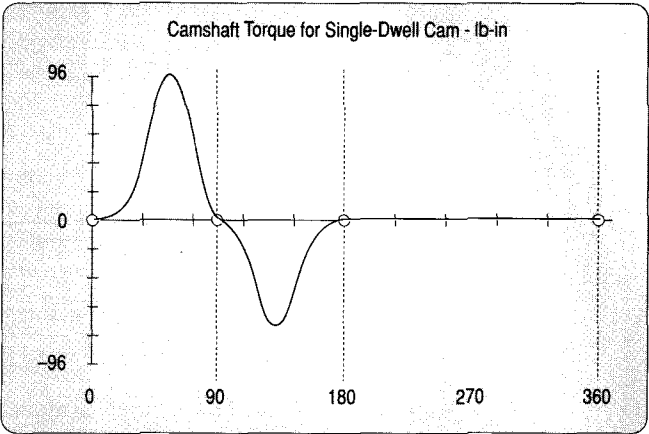


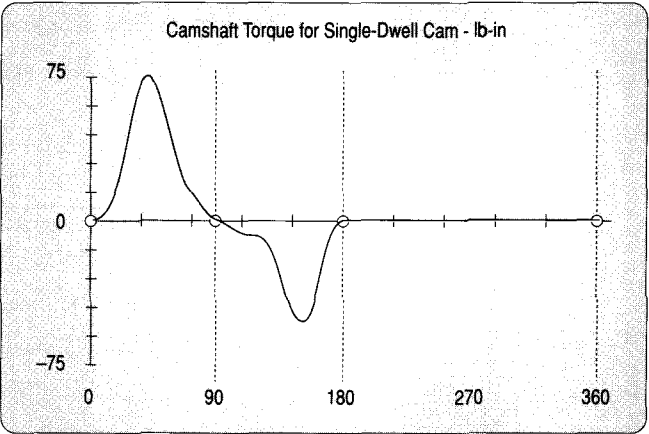
FIGURE 15-14

Dynamic forces in three different designs of a single-dwell cam

(a) Double harmonic rise -
double harmonic fall



(b) Cycloidal rise -
cycloidal fall



(c) Sixth-degree
polynomial

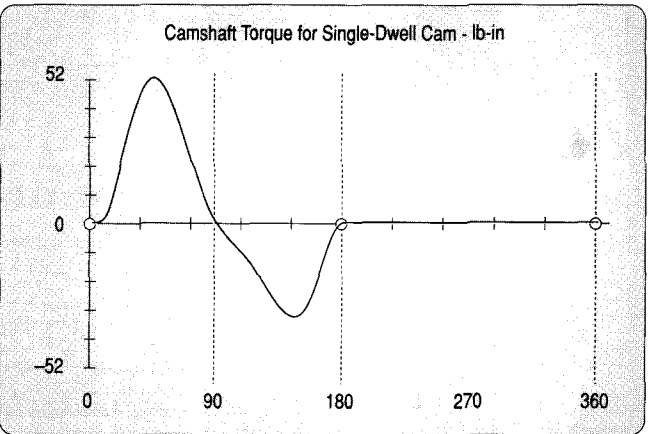
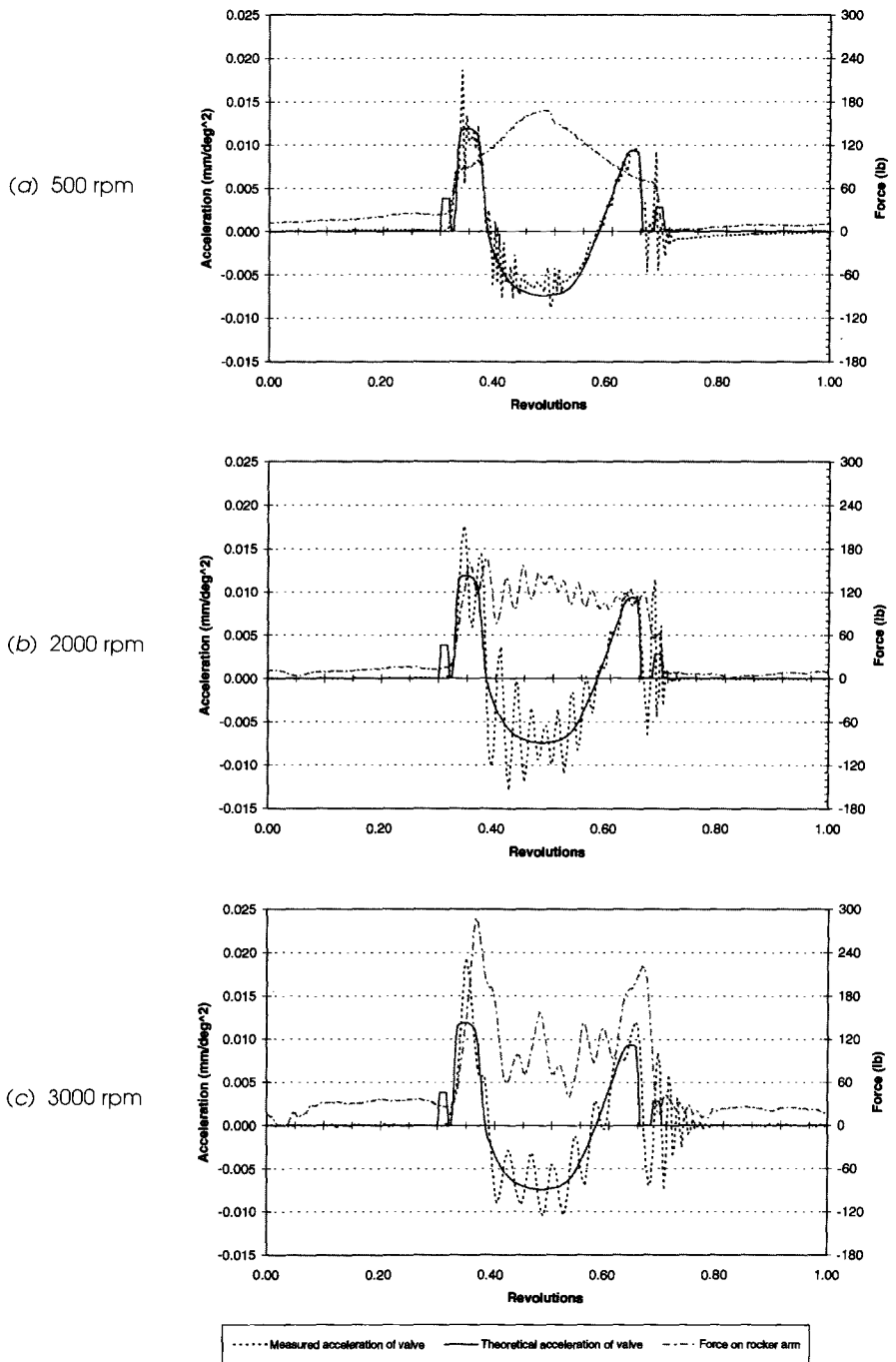


FIGURE 15-15
Dynamic input torque in three different designs of a single-dwell cam

**FIGURE 15-16**

Valve acceleration and rocker arm force in a single overhead cam valve train

quency of the valve spring at about 2000 camshaft rpm, causing the spring to resonate. The same effects can be seen in the rocker arm force. Everything in a machine tends to sympathetically vibrate at its own natural frequency when excited by any input forcing function. Sensitive transducers such as accelerometers will pick up these vibrations as they are transmitted through the structure.

15.7 PRACTICAL CONSIDERATIONS

Koster^[1] proposes some general rules for the design of cam-follower systems for high-speed operation based on his extensive dynamic modeling and experimentation.

To minimize the positional error and residual acceleration:

- 1 Keep the total lift of the follower to a minimum.
- 2 If possible, arrange the follower spring to preload all pivots in a consistent direction to control backlash in the joints.
- 3 Keep the duration of rises and falls as long as possible to minimize τ .
- 4 Keep follower train mass low and follower train stiffness high to increase natural frequency.
- 5 Any lever ratios present will change the effective stiffness of the system by the square of the ratio. Try to keep lever ratios close to 1.
- 6 Make the camshaft as stiff as possible **both in torsion and in bending**. This is perhaps the most important factor in controlling follower vibration.
- 7 Reduce pressure angle by increasing the cam pitch circle diameter.
- 8 Use low backlash or antibacklash gears in the camshaft drive train.

15.8 REFERENCES

- 1 **Koster, M. P.** (1974). *Vibrations of Cam Mechanisms*. Phillips Technical Library Series, Macmillan Press Ltd.: London.
- 2 **Norton, R. L., et al.** (1998). "Analyzing Vibrations in an IC Engine Valve Train." SAE Paper: 980570.

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- Rothbart, H. A.** (1956). *Cams: Design, Dynamics, and Accuracy*. John Wiley & Sons: New York, p. 350.

15.10 PROBLEMS

Program DYNACAM may be used to solve these problems where applicable. Where units are unspecified, work in any consistent units system you wish. Appendix D contains some pages from a catalog of commercially available helical coil springs to aid in designing realistic solutions to these problems.

- *†15-1 Design a double-dwell cam to move a follower of mass = 2.2 bl from 0 to 2.5 inches in 60° with modified sine acceleration, dwell for 120°, fall 2.5 inches in 30° with cycloidal motion, and dwell for the remainder. The total cycle must take 4 sec. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.2 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- *†15-2 Design a double-dwell cam to move a follower of mass = 1.4 bl from 0 to 1.5 inches in 45° with 3-4-5 polynomial motion, dwell for 150°, fall 1.5 inches in 90° with 4-5-6-7 polynomial motion, and dwell for the remainder. The total cycle must take 6 sec. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.1 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- *†15-3 Design a single-dwell cam to move a follower of mass = 3.2 bl from 0 to 2 inches in 60°, fall 2 inches in 90°, and dwell for the remainder. The total cycle must take 5 sec. Use a seventh-degree polynomial. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.15 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- *†15-4 Design a three-dwell cam to move a follower of mass = 0.4 bl from 0 to 2.5 inches in 40°, dwell for 100°, fall 1.5 inches in 90°, dwell for 20°, fall 1 inch in 30°, and dwell for the remainder. The total cycle must take 10 sec. Choose suitable programs for rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.12 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM which is on the attached CD-ROM.

- *†15-5 Design a four-dwell cam to move a follower of mass = 1.25 bl from 0 to 2.5 inches in 40° , dwell for 100° , fall 1.5 inches in 90° , dwell for 20° , fall 0.5 inches in 30° , dwell for 40° , fall 0.5 inches in 30° , and dwell for the remainder. The total cycle must take 15 sec. Choose suitable programs for rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.18 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- *†15-6 A mass-spring damper system as shown in Figure 15-1b (p. 687) has the values shown in Table P15-1. Find the undamped and damped natural frequencies and the value of critical damping for the system(s) assigned.
- †15-7 Figure P15-1 shows a cam-follower system. The dimensions of the solid, rectangular 2×2.5 in cross-section aluminum arm are given. The cutout for the 2-in-diameter, 1.5-in-wide steel roller follower is 3 in long. Find the arm's mass, center of gravity location, and mass moment of inertia about both its CG and the arm pivot. Create a linear, one-DOF lumped mass model of the dynamic system referenced to the cam-follower and calculate the cam-follower force for one revolution. The cam is a pure eccentric with eccentricity = 0.5 in and turns at 500 rpm. The spring has a rate of 123 lb/in and a preload of 173 lb. Ignore damping.
- †‡15-8 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 2.5 inches in 60° with modified sine acceleration, dwell for 120° , fall 2.5 inches in 30° with cycloidal motion, and dwell for the remainder. Cam speed is 100 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.10.
- †‡15-9 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 1.5 inches in 45° with 3-4-5 polynomial motion, dwell for 150° , fall 1.5 inches in 90° with 4-5-6-7 polynomial motion, and dwell for the remainder. Cam speed is 250 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.

TABLE P15-1
Problem 15-6

	<i>m</i>	<i>k</i>	<i>c</i>
a.	1.2	14	1.1
b.	2.1	46	2.4
c.	30.0	2	0.9
d.	4.5	25	3.0
e.	2.8	75	7.0
f.	12.0	50	14.0

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program *DYNACAM* which is on the attached CD-ROM.

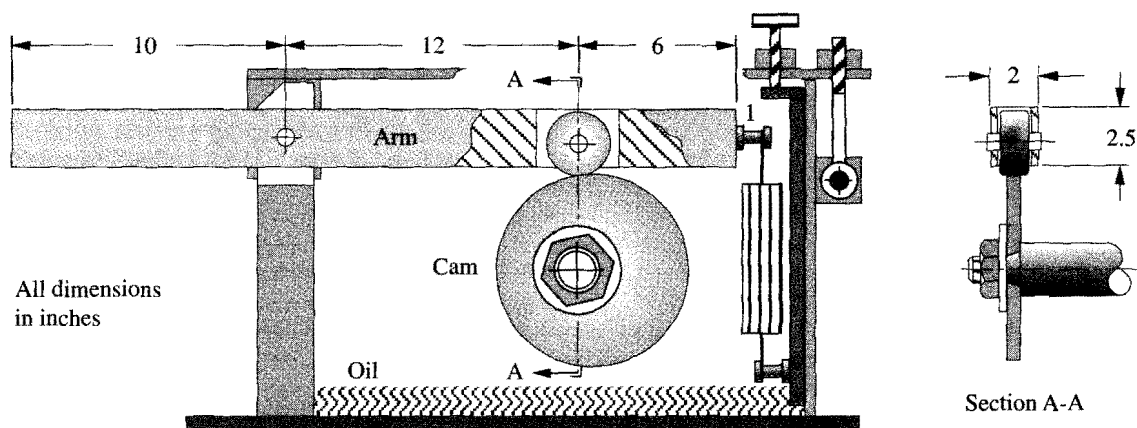
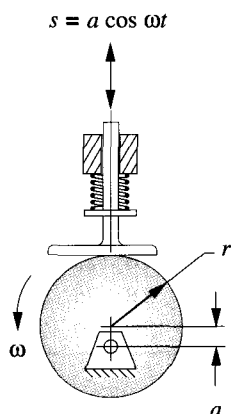


FIGURE P15-1

Problems 15-7 to 15-11

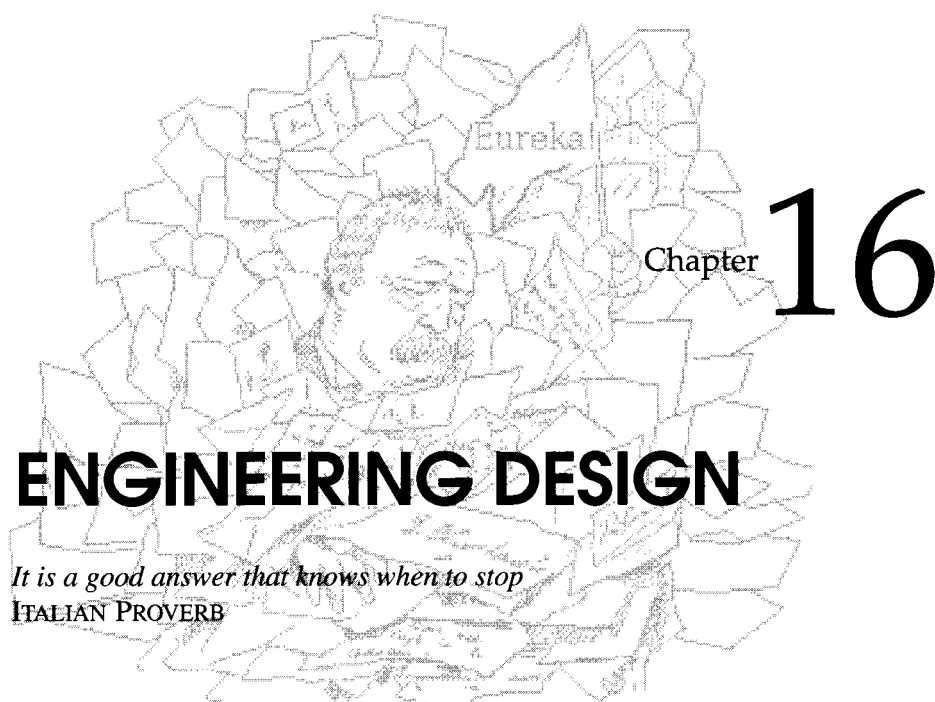
**FIGURE P15-2**

Problems 15-12 to 15-14

- †‡15-10 Repeat Problem 15-7 for a single-dwell cam to move the follower from 0 to 2 inches in 60° , fall 2 inches in 90° , and dwell for the remainder. Use a seventh-degree polynomial. Cam speed is 250 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.
- †‡15-11 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 2 inches in 45° with cycloidal motion, dwell for 150° , fall 2 inches in 90° with modified sine motion, and dwell for the remainder. Cam speed is 200 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.
- †‡15-12 The cam in Figure P15-2 is a pure eccentric with eccentricity = 20 mm and turns at 200 rpm. The mass of the follower is 1 kg. The spring has a rate of 10 N/m and a preload of 0.2 N. Find the follower force over one revolution. Assume a damping ratio of 0.10. If there is follower jump, respecify the spring rate and preload to eliminate it.
- †‡15-13 Repeat Problem 15-12 using a cam with a 20-mm symmetric double harmonic rise and fall (180° rise - 180° fall). See Chapter 8 for cam formulas.
- ‡15-14 Repeat Problem 15-12 using a cam with a 20-mm 3-4-5-6 polynomial rise and fall (180° rise - 180° fall). See Chapter 8 for cam formulas.
- ‡15-15 Design a double-dwell cam to move a follower of mass = 2 kg from 0 to 45 mm in 60° with modified sine acceleration, dwell for 120° , fall 45 mm in 90° with 3-4-5 polynomial motion, and dwell for the remainder. The total cycle must take 1 sec. Size a return spring and specify its preload to maintain contact between cam and follower. Select a spring from Appendix D. Calculate and plot the dynamic force and torque. Assume damping of 0.25 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- ‡15-16 Design a single-dwell cam using polynomials to move a follower of mass = 10 kg from 0 to 25 mm in 60° , fall 25 mm in 90° , and dwell for the remainder. The total cycle must take 2 sec. Size a return spring and specify its preload to maintain contact between cam and follower. Select a spring from Appendix D. Calculate and plot the dynamic force and torque. Assume damping of 0.15 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- ‡15-17 Design a four-dwell cam to move a follower of mass = 3 kg from 0 to 40 mm in 40° , dwell for 100° , fall 20 mm in 90° , dwell for 20° , fall 10 mm in 30° , dwell for 40° , fall 10 mm in 30° , and dwell for the remainder. The total cycle must take 10 sec. Choose suitable programs for rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.25 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.
- ‡15-18 Design a cam to drive an automotive valve train whose effective mass is 0.2 kg. $\zeta = 0.3$. Valve stroke is 12 mm. Roller follower is 10 mm diameter. The open-close event occupies 160° of camshaft revolution; dwell for remainder. Use one or two polynomials for the rise-fall event. Select a spring constant and preload to avoid jump to 3500 rpm. Fast opening and closing and maximum open time are desired.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM which is on the attached CD-ROM.



ENGINEERING DESIGN

It is a good answer that knows when to stop
ITALIAN PROVERB

16.0 INTRODUCTION

Of all the myriad activities that the practicing engineer engages in, the one that is at once the most challenging and potentially the most satisfying is that of design. Doing calculations to analyze a clearly defined and structured problem, no matter how complex, may be difficult, but the exercise of creating something from scratch, to solve a problem that is often poorly defined, is very difficult. The sheer pleasure and joy at conceiving a viable solution to such a design problem is one of life's great satisfactions for anyone, engineer or not. The preceding chapters have attempted to present the particular subject matter in a way and context that will not only enhance the student's fundamental understanding of the topic but also encourage and promote his or her creative efforts toward the solution of design problems.

As a closing to a book that began with a discussion of a design process in Chapter I, it is perhaps appropriate to present something of a "case study" of the design process to conclude the discussion. Some years ago, a very creative engineer of the author's acquaintance, George A. Wood Jr., heard a presentation by another creative engineer of the author's acquaintance, Keivan Towfigh, about one of his designs. Years later, Mr. Wood himself wrote a short paper about creative engineering design in which he reconstructed Mr. Towfigh's presumed creative process when designing the original invention. Both Mr. Wood and Mr. Towfigh have kindly consented to the reproduction of that paper here. It serves, in this author's opinion, as an excellent example and model for the student of engineering design to consider when pursuing his or her own design career.

16.1 A DESIGN CASE STUDY

Educating for Creativity in Engineering[1]

by GEORGE A. WOOD JR.

*One facet of engineering, as it is practiced in industry, is the creative process. Let us define creativity as Rollo May does in his book, *The Courage to Create*.¹ It is "the process of bringing something new into being." Much of engineering has little to do with creativity in its fullest sense. Many engineers choose not to enter into creative enterprise, but prefer the realms of analysis, testing and product or process refinement. Many others find their satisfaction in management or business roles and are thus removed from engineering creativity as we shall discuss it here.*

From the outset, I wish to note that the less creative endeavors are no less important or satisfying to many engineers than is the creative experience to those of us with the will to create. It would be a false goal for all engineering schools to assume that their purpose was to make all would-be engineers creative and that their success should be measured by the "creative quotient" of their graduates.

On the other hand, for the student who has a creative nature, a life of high adventure awaits if he can find himself in an academic environment which recognizes his needs, enhances his abilities and prepares him for a place in industry where his potential can be realized.

In this talk I will review the creative process as I have known it personally and witnessed it in others. Then I shall attempt to indicate those aspects of my training that seemed to prepare me best for a creative role and how this knowledge and these attitudes toward a career in engineering might be reinforced in today's schools and colleges.

During a career of almost thirty years as a machine designer, I have seen and been a part of a number of creative moments. These stand as the high points of my working life. When I have been the creator I have felt great elation and immense satisfaction. When I have been with others at their creative moments I have felt and been buoyed up by their delight. To me, the creative moment is the greatest reward that the profession of engineering gives.

*Let me recount an experience of eight years ago when I heard a paper given by a creative man about an immensely creative moment. At the First Applied Mechanisms Conference in Tulsa, Oklahoma, was a paper entitled *The Four-Bar Linkage as an Adjustment Mechanism*.² It was nestled between two "how to do it" academic papers with graphs and equations of interest to engineers in the analysis of their mechanism problems. This paper contained only one very elementary equation and five simple illustrative figures; yet, I remember it now more clearly than any other paper I have ever heard at mechanism conferences. The author was Keivan Towfigh and he described the application of the geometric characteristics of the instant center of the coupler of a four bar mechanism.*

His problem had been to provide a simple rotational adjustment for the oscillating mirror of an optical galvanometer. To accomplish this, he was required to rotate the entire galvanometer assembly about an axis through the center of the mirror and perpendicular to the pivot axis of the mirror. High rigidity of the system after adjustment was essential with very limited space available and low cost required, since up to sixteen of these galvanometer units were used in the complete instrument.

His solution was to mount the galvanometer elements on the coupler link of a one-piece, flexure hinged, plastic four bar mechanism so designed that the mirror center was at the instant center of the linkage at the midpoint of its adjustment. (See Fig 4.) It is about this particular geometric point (see Fig 1.) That pure rotation occurs and with proper selection of linkage dimensions this condition of rotation without translation could be made to hold sufficiently accurately for the adjustment angles required.

Unfortunately, this paper was not given the top prize by the judges of the conference. Yet, it was, indirectly, a description of an outstandingly creative moment in the life of a creative man.

Let us look at this paper together and build the steps through which the author probably progressed in the achievement of his goal. I have never seen Mr. Towfigh since, and I shall therefore describe a generalized creative process which may be incorrect in some details but which, I am sure, is surprisingly close to the actual story he would tell.

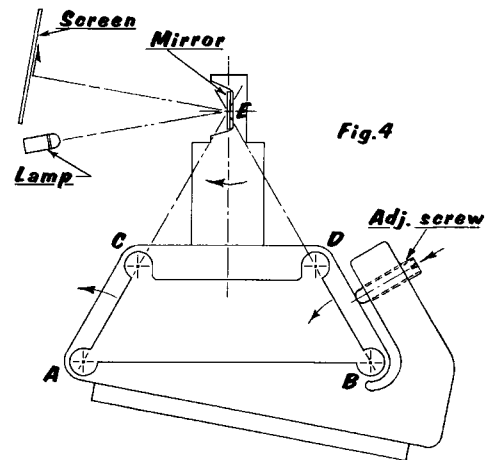
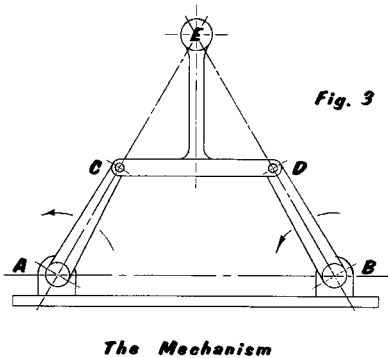
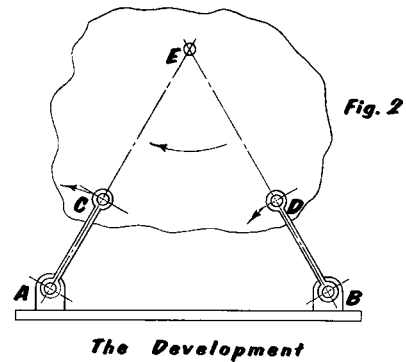
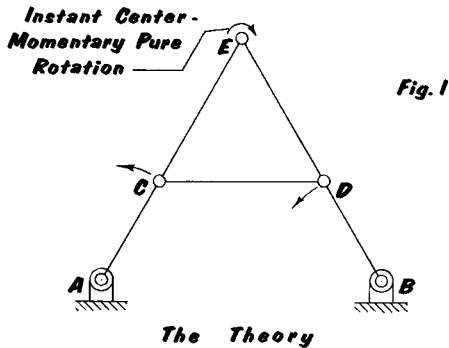
The galvanometer problem was presented to Mr. Towfigh by his management. It was, no doubt, phrased something like this: "In our new model, we must improve the stability of the adjustment of the equipment but keep the cost down. Space is critical and low weight is too. The overall design must be cleaned up, since customers like modern, slim-styled equipment and we'll lose sales to others if we don't keep ahead of them on all points. Our industrial designer has this sketch that all of us in sales like and within which you should be able to make the mechanism fit."

Then followed a list of specifications the mechanism must meet, a time when the new model should be in production and, of course, the request for some new feature that would result in a strong competitive edge in the marketplace.

I wish to point out that the galvanometer adjustment was probably only one hoped-for improvement among many others. The budget and time allowed were little more than enough needed for conventional redesign, since this cost must be covered by the expected sales of the resulting instrument. For every thousand dollars spent in engineering, an equivalent increase in sales or reduction in manufacturing cost must be realized at a greater level than the money will bring if invested somewhere else.

In approaching this project, Mr. Towfigh had to have a complete knowledge of the equipment he was designing. He had to have run the earlier models himself. He must have adjusted the mirrors of existing machines many times. He had to be able to visualize the function of each element in the equipment in its most basic form.

(research)



Secondly, he had to ask himself (as if he were the customer) what operational and maintenance requirements would frustrate him most. He had to determine which of these might be improved within the design time available. In this case he focused on the mirror adjustment. He considered the requirement of rotation without translation. He determined the maximum angles that would be necessary and the allowable translation that would not affect the practical accuracy of the equipment. He recognized the desirability of a one screw adjustment. He spent a few hours thinking of all the ways he had seen of rotating an assembly about an arbitrary point. He kept rejecting each solution as it came to him as he felt, in each case, that there was a better way. His ideas had too many parts, involved slides, pivots, too many screws, were too vibration sensitive or too large.

He thought about the problem that evening and at other times while he proceeded with the design of other aspects of the machine. He came back to the problem several times during the next few days. His design time was running out. He was a mechanism specialist and visualized a host of cranks and bars moving the mirrors. Then one day, probably after a period when he had turned his attention elsewhere, on rethinking of the

adjustment device an image of the system based on one of the elementary characteristics of a four bar mechanism came to him.

I feel certain that this was a visual image, as clear as a drawing on paper. It was probably not complete but involved two inspirations. First was the characteristics of the instant center. (See Figs 1,2,3.) Second was the use of flexure hinge joints which led to a one-piece plastic molding. (See Fig 4.) I am sure that at this moment he had a feeling that this solution was right. He knew it with certainty. The whole of his engineering background told him. He was elated. He was filled with joy. His pleasure was not because of the knowledge that his superiors would be impressed or that his security in the company would be enhanced. It was the joy of personal victory, the awareness that he had conquered.

The creative process has been documented before by many others far more qualified to analyze the working of the human mind than I. Yet I would like to address, for the remaining minutes, how education can enhance this process and help more engineers, designers and draftsmen extend their creative potential.

The key elements I see in creativity that have greatest bearing on the quality that results from the creative effort are visualization and basic knowledge that gives strength to the feeling that the right solution has been achieved. There is no doubt in my mind that the fundamental mechanical principles that apply in the area in which the creative effort is being made must be vivid in the mind of the creator. The words that he was given in school must describe real elements that have physical, visual significance. $F = ma$ must bring a picture to his mind vivid enough to touch.

If a person decides to be a designer, his training should instill in him a continuing curiosity to know how each machine he sees works. He must note its elements and mentally see them function together even when they are not moving. I feel that this kind of solid, basic knowledge couples with physical experience to build ever more critical levels at which one accepts a tentative solution as "right."

(analysis)

It should be noted that there have been times for all of us when the inspired "right" solution has proven wrong in the long run. That this happens does not detract from the process but indicates that creativity is based on learning and that failures build toward a firmer judgment base as the engineer matures. These failure periods are only negative, in the growth of a young engineer, when they result in the fear to accept a new challenge and begot excessive caution which then stifles the repetition of the creative process.

What would seem the most significant aspects of an engineering curriculum to help the potentially creative student develop into a truly creative engineer?

First is a solid, basic knowledge in physics, mathematics, chemistry and those subjects relating to his area of interest. These fundamentals should have physical meaning to the student and a vividness that permits him to explain his thoughts to the untrained layman. All too often technical words are used to cover cloudy concepts. They serve the ego of the user instead of the education of the listener.

Second is the growth of the student's ability to visualize. The creative designer must be able to develop a mental image of that which he is inventing. The editor of the book *Seeing with the Mind's Eye*⁴¹ by Samuels and Samuels, says in the preface:

"... visualization is the way we think. Before words, images were. Visualization is the heart of the bio-computer. The human brain programs and self-programs through its images. Riding a bicycle, driving a car, learning to read, baking a cake, playing golf - all skills are acquired through the image making process. Visualization is the ultimate consciousness tool."

Obviously, the creator of new machines or products must excel in this area.

To me, a course in Descriptive Geometry is one part of an engineer's training that enhances one's ability to visualize theoretical concepts and graphically reproduce the result. This ability is essential when one sets out to design a piece of new equipment. First, he visualizes a series of complete machines with gaps where the problem or unknown areas are. During this time, a number of directions the development could take begin to form. The best of these images are recorded on paper and then are reviewed with those around him until, finally, a basic concept emerges.

The third element is the building of the student's knowledge of what can be or has been done by others with different specialized knowledge than he has. This is the area to which experience will add throughout his career as long as he maintains an enthusiastic curiosity. Creative engineering is a building process. No one can develop a new concept involving principles about which he has no knowledge. The creative engineer looks at problems in the light of what he has seen, learned and experienced and sees new ways for combining these to fill a new need.

Fourth is the development of the ability of the student to communicate his knowledge to others. This communication must involve not only skills with the techniques used by technical people but must also include the ability to share engineering concepts with untrained shop workers, business people and the general public. The engineer will seldom gain the opportunity to develop a concept if he cannot pass on to those around him his enthusiasm and confidence in the idea. Frequently, truly ingenious ideas are lost because the creator cannot transfer his vivid image to those who might finance or market it.

Fifth is the development of a student's knowledge of the physical result of engineering. The more he can see real machines doing real work, the more creative he can be as a designer. The engineering student should be required to run tools, make products, adjust machinery and visit factories. It is through this type of experience that judgement grows as to what makes a good machine, when approximation will suffice and where optimization should halt.

It is often said that there has been so much theoretical development in engineering during the past few decades that the colleges and universities do not have time for the basics I have outlined above. It is suggested that industry should fill in the practice areas that colleges have no time for, so that the student can be exposed to the latest technology. To some degree I understand and sympathize with this approach, but I feel that there is a negative side that needs to be recognized. If a potentially creative engineer leaves col-

lege without the means to achieve some creative success as he enters his first job, his enthusiasm for creative effort is frustrated and his interest sapped long before the most enlightened company can fill in the basics. Therefore, a result of the "basics later" approach often is to remove from the gifted engineering student the means to express himself visually and physically. Machine design tasks therefore become the domain of the graduates of technical and trade schools and the creative contribution by many a brilliant university student to products that could make all our lives richer is lost.

As I said at the start, not all engineering students have the desire, drive and enthusiasm that are essential to creative effort. Yet I feel deeply the need for the enhancement of the potential of those who do. That expanding technology makes course decisions difficult for both student and professor is certainly true. The forefront of academic thought has a compelling attraction for both the teacher and the learner. Yet I feel that the development of strong basic knowledge, the abilities to visualize, to communicate, to respect what has been done, to see and feel real machinery, need not exclude or be excluded by the excitement of the new. I believe that there is a curriculum balance that can be achieved which will enhance the latent creativity in all engineering and science students. It can give a firm basis for those who look towards a career of mechanical invention and still include the excitement of new technology.

I hope that this discussion may help in generating thought and providing some constructive suggestions that may lead more engineering students to find the immense satisfaction of the creative moment in the industrial environment. In writing this paper I have spent considerable time reflecting on my years in engineering and I would close with the following thought. For those of us who have known such times during our careers, the successful culminations of creative efforts stand among our most joyous hours.

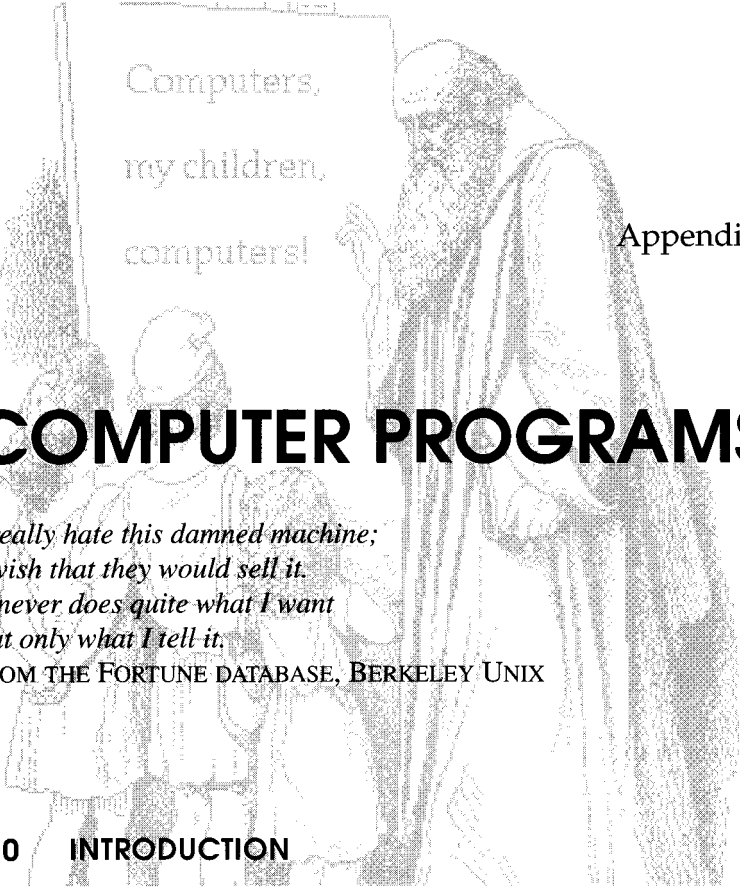
16.2 CLOSURE

Mr. Wood's description of his creative experiences in engineering design and the educational factors which influenced them closely parallel this author's experience as well. The student is well advised to follow his prescription for a thorough grounding in the fundamentals of engineering and communication skills. A most satisfying career in the design of machinery can result.

16.3 REFERENCES

1. **Wood, G. A. Jr.** (1977). "Educating for Creativity in Engineering," Presented at the 85th Annual Conf. ASEE, Univ. of No. Dakota.
2. **May, R.** (1976). *The Courage to Create*, Bantam Books, New York.
3. **Towfigh, K.** (1969). "The Four-Bar Linkage as an Adjustment Mechanism," 1st Applied Mechanism Conf., Oklahoma State Univ., Tulsa Okla., pp. 27-1- 27-4.
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Computers,
my children,
computers!

Appendix

A

COMPUTER PROGRAMS

*I really hate this damned machine;
I wish that they would sell it.
It never does quite what I want
But only what I tell it.*

FROM THE FORTUNE DATABASE, BERKELEY UNIX

A.0 INTRODUCTION

In addition to the student version of the commercial simulation program *Working Model*, there are seven custom computer programs provided on the CD-ROM with this text: programs FOURBARFIVEBAR, SIXBAR, SLIDER, MATRIX, DYNACAM, and ENGINE. These are student editions of the programs. Professional versions with extended capabilities are also available. Contact the author at rlnorton@wpi.edu or visit his faculty page at the web site <http://www.wpi.edu> for more information. Programs FOURBAR, FIVEBAR, SIXBAR, and SLIDER are based on the mathematics derived in Chapters 4 to 7 and 10 to 11 and use the equations presented therein to solve for position, velocity, and acceleration in linkages of the variety described in the particular program's name. Program DYNACAM is a beam design program based on the mathematics derived in Chapters 8 and 15. Program ENGINE is based on the mathematics derived in Chapters 13 and 14. Program MATRIX is a general linear simultaneous equation solver. All have similar choices for display of output data in the form of tables and plots. All the programs are designed to be user friendly and reasonably "crashproof." The author encourages users to email reports of any "bugs" or problems encountered in their use to him at rlnorton@wpi.edu.

Learning Tools

All the custom programs provided with this text are designed to be learning tools to aid in the understanding of the relevant subject matter and *are specifically not intended to*

be used for commercial purposes in the design of hardware and should not be so used. It is quite possible to obtain inappropriate (but mathematically correct) results to any problem solved with these programs, due to incorrect or inappropriate input of data. In other words, the user is expected to understand the kinematic and dynamic theory underlying the program's structure and to also understand the mathematics on which the program's algorithms are based. This information on the underlying theory and mathematics is derived and described in the noted chapters of this text. Most equations used in the programs are presented or derived in this textbook.

Disclaimer

Commercial software for use in design or analysis needs to have built-in safeguards against the possibility of the user providing incorrect, inappropriate, or ridiculous values for input variables, in order to guard against erroneous results due to user ignorance or inexperience. **The student editions of the custom programs provided with this text are not commercial software and deliberately do not contain such safeguards against improper input data,** on the premise that to do so would "short circuit" the student's learning process. We learn most from our failures. These programs provide a consequence-free environment to explore failure of your designs "on paper" and in the process come to a more thorough and complete understanding of the subject matter. **The author and publisher are not responsible for any damages which may result from the use or misuse of these programs.**

Brute Force and Ignorance

The very rapid computation speed of these programs allows the student to explore a much larger number and variety of potential solutions to more realistic and comprehensive problems than could be accomplished using only hand calculator solutions of these complicated systems of equations. This is both an advantage and a danger. The advantage is that the student can use the programs like a "flight simulator" to "fly" potential design solutions through their paces with no consequences from a "crash" of the design not yet built. If the student diligently attempts to interpret the program's results and relate them to the relevant theory, a more thorough understanding of the kinematics and dynamics can result. On the other hand, there is a great temptation to use these programs with "brute force and ignorance" (BFI) to somewhat randomly try solutions without regard to what the theory and equations are telling you and hope that somehow a usable solution will "pop out." The student who succumbs to this temptation will not obtain much benefit from the exercise and will probably have a poor design result. This situation is probably best summed up in the following comment from a student who had suffered through the course in kinematics using these programs to design solutions to three project problems like those listed at the end of Chapter 3.

... The computer, with its immense benefits for the engineer, can also be a hindrance if one does not first develop a thorough understanding of the theory upon which a particular program is based. An over-reliance on the computer can leave one "computer smart" and "engineering stupid." The BFI approach becomes increasingly tempting when it can be employed with such ease *BRIAN KIMBALL*

Smart student! Use these computer programs wisely. Avoid *Brute Force* and *Ignorance*! **Engineer** your solutions and understand the theory behind them.

A.1 GENERAL INFORMATION

Hardware Requirements

These programs work on any computer that runs Windows 3.1, Windows 95, or Windows NT. Different versions of the programs are provided for each operating system. A CD-ROM drive is needed, as is a hard disk drive. If your computer does not read a CD-ROM, your school's computer center can probably convert it to a usable format for you.

Operating System Requirements

These custom programs are written in Microsoft Visual Basic and are compiled in two versions: *Pgmname16.exe* for the Windows 3.1 (16 bit) operating system and *Pgmname32.exe* for the Windows 95 and NT (32 bit) operating systems. Be sure to use the version suitable to your computer's hardware and software.

Installing the Software

The CD-ROM contains the executable program files plus all necessary Dynamic Link Library (DLL) and other ancillary files needed to run the programs. Run the SETUP file from the individual program's folder on the CD-ROM to automatically decompress and install all of its files on your hard drive. A folder and icon will be created with that program's name. The program can then be run by double clicking on the icon or on the *Pgmname16.exe* or *Pgmname32.exe* file itself. In Windows 95 or NT the program name will appear in the list under the *Start* button's *Program* menu after installation and can be run from there.

How to Use This Manual

This manual is intended to be used while running the programs. To see a screen referred to, bring it up within the program to follow its discussion.

A.2 GENERAL PROGRAM OPERATION

All seven programs in the set have similar features and operate in a consistent way. For example, all printing and plotting functions are selected from identical screens common to all programs. Opening and saving files are done identically in all programs. These common operations will be discussed in this section independent of the particular program. Later sections will address the unique features and operations of each program.

Note that *student editions* of these programs are supplied with this book at no charge and carry a limited-term license restricted to use in course work for up to two semesters (9 months). If you continue to use the program after that time, you are expected to register it and pay the shareware fee defined on the *Registration* screen. If you wish to use the program for the benefit of a company or for any commercial purpose, then you must obtain the professional edition of the same program. The student editions are not to be used commercially. The professional editions typically offer more features than the student editions.

Running the Programs (All Programs)

At start-up, a splash screen appears which identifies the program version, revision number, and revision date. Click the button labeled *Start* or press the *Enter* key to run the program. A *Disclaimer* screen next appears which defines the registered owner and allows the printing of a registration form if the software is as yet unregistered. A registration form can be accessed and printed from this screen.

The next screen, the *Title* screen, allows the input of any user and/or project identification desired. This information must be provided to proceed and is used to identify all plots and printouts from this program session. The second box on the *Title* screen allows any desired file name to be supplied for storing data to disk. This name defaults to **Modell** and may be changed at this screen and/or when later writing the data to disk. The third box allows the typing of a starting design number for the first design. This design number defaults to 1 and is automatically incremented each time you change the basic design during this program session. It is used only to identify plots, data files, and printouts so they can be grouped, if necessary, at a later date. When the *Done* button on the *Title* screen is clicked, the *Home* screen appears.

The Home Screen (All Programs)

All program actions start and end at the *Home* screen which has several pull-down menus and buttons, some of which commands (*File*, *New*, *Open*, *Save*, *Save As*, *Units*, *About*, *Plot*, *Print*, *Quit*) are common to all programs. These will be described below.

General User Actions Possible Within a Program (All Programs)

The programs are constructed to allow operation from the keyboard or the mouse or with any combination of both input devices. Selections can be made either with the mouse or, if a button is highlighted (showing a dotted square within the button), the *Enter* key will activate the button as if it had been clicked with the mouse. Text boxes are provided where you need to type in data. These have a yellow background. In general, what you type in any text box is not accepted until you hit the *Enter* key or move off that box with the *Tab* key or the mouse. This allows you to retype or erase with no effect until you leave the text box. You can move between available input fields with the *Tab* key (and backup with *Shift-Tab*) on most screens. If you are in doubt as to the order in which to input the data on any screen, try using the *Tab* key as it will take you to each needed entry field in a sensible order. You can then type or mouse click to input the desired data in that field. Remember that a yellow background means typed input data is expected. Boxes with a cyan background provide information back to you but cannot be typed in.

Other information required from you is selected from drop-down menus or lists. These have a white background. Some lists allow you to type in a value different than any provided in the available list of selections. If you type an inappropriate response, it will simply ignore you or choose the closest value to your request. Typing the first few letters of a listed selection will sometimes cause it to be selected. Double clicking on a selectable item in a list will often provide a shortcut.

Units (All Programs)

The *Units* menu defines several units systems to choose from. *Note that all programs work entirely in pure numbers without regard to units.* It is your responsibility to ensure that the data as input are in some consistent units system. No units conversion is done within the programs. The *Units* menu selection that you make has only one effect, namely to change the labels displayed on various input and output parameters within the program. You mix units systems at your own peril.

Examples (Most Programs)

Most of the programs have an *Examples* pull-down menu on the *Home* screen which provides some number of example mechanisms that will demonstrate the program's capability. Selecting an example from this menu will cause calculation of the mechanism and open a screen to allow viewing the results. In some cases you may need to hit a button marked *Calculate*, *Run*, or *Animate* on the presented screen to see the results. Some programs also provide access to examples from various screens.

Creating New, Saving, and Opening Files (File - All Programs)

The standard *Windows* functions for creating new files, saving your work to disk, and opening a previously saved file are all accessible from the pull-down menu labeled *File* on each program's *Home* screen. Selecting *New* from this menu will zero any data you may have already created within the program, but before doing so will give warning and prompt you to save the data to disk.

The *Save* and *Save As* selections on the *File* menu prompt you to provide a file name and disk location to save your current model data to disk. The data are saved in a custom format and with a three-character suffix unique to the particular program. You should use the recommended suffix on these files as that will allow the program to see them on the disk when you want to open them later. If you forget to add the suffix when saving a file, you can still recover the file.

Selecting *Open* from the *File* menu prompts you to pick a file from those available in the disk directory that you choose. If you do not see any files with the program's suffix, use the pull-down menu within the *Open File* dialog box to choose *Show All Files* and you will then see them. They will read into the program properly with or without the suffix in their name as long as they were saved from the same program. If you are a former user of the DOS versions of these programs, files saved from the older version of a given program can also be opened in its *Windows* version, though they will not have as much information as is needed in the new programs.

Copying Screens to Clipboard or Printer (Copy - All Programs)

Any screen can be copied as a graphic to the clipboard by using the standard *Windows* keyboard combo of *Alt-PrintScrn*. It will then be available for pasting into any compatible *Windows* program such as *Word* or *Powerpoint* that is running concurrently in *Windows*. Most screens also provide a button to dump the screen image to an attached laser printer. However, the quality of that printed image may be less than could be obtained

by copying and pasting the image into a program such as *Word* or *Powerpoint* and then printing it from that program. It seems that *Visual Basic* does not print graphics as well as some other *Windows* applications.

Printing to Screen, Printer, and Exporting Disk Files (*Print* Button)

Selecting the *Print* button from the *Home* screen will open the *Print Select* screen (see Figure A-1) containing lists of variables that may be printed. Buttons on the left of this screen can be clicked to direct the printed output to one of *Screen*, *Printer*, or *Disk*. This choice defaults to *Screen* and so must be clicked each time the screen is opened to obtain either of the other options. The output is different with each of these selections.

Selecting *Screen* will result in a scrollable screen window full of the requested data. Scrolling will allow you to view all data requested serially. This data screen can be copied to the clipboard or dumped to a printer as described above, but this clip or dump will typically show only a portion of the available data, i.e., one screenful.

Selecting *Printer* as the output device will cause the entire selection of data to print to an available printer. Only some of the sidebar information shown on the screen display will be included in this printout.

Selecting *Disk* as the output device will cause your selections to be sent to the file of your choice in an ASCII text format (tab delimited) that can be opened in a spreadsheet program such as *Lotus 123* or *Excel*. You can then do further calculations or plotting of data within the spreadsheet program.

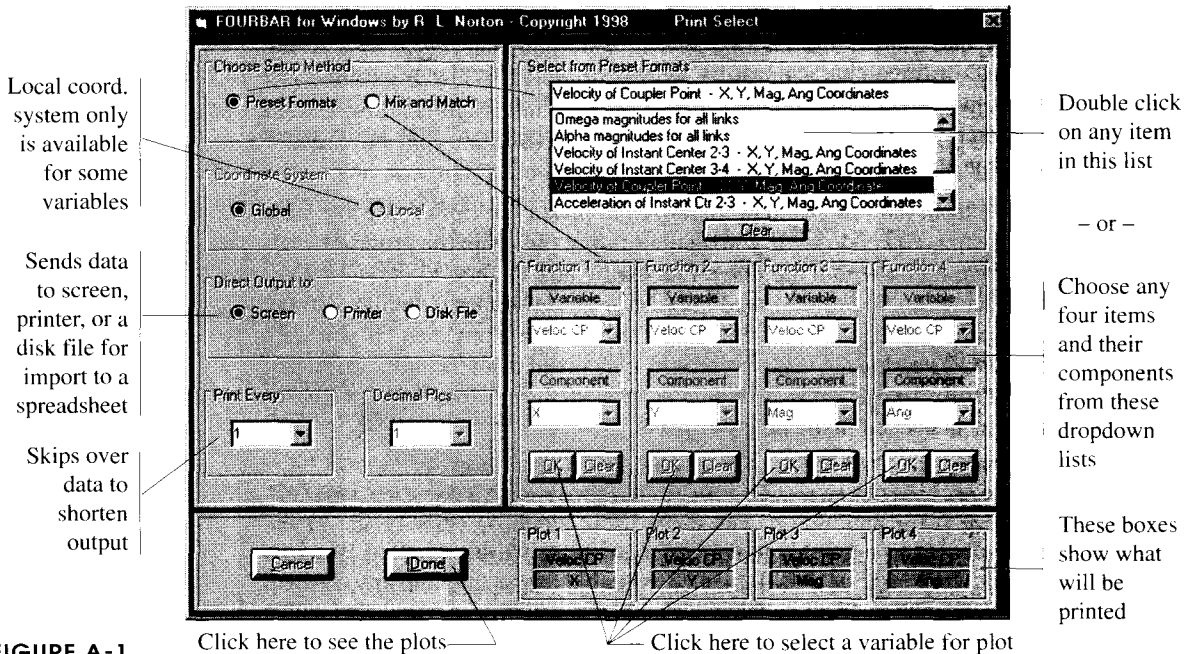


FIGURE A-1

The *Print Select* screen is common to all programs (not all programs allow component selection)

The *Print Select* screen has two modes for data selection, *Preset Formats* and *Mix and Match*. The former provides preselected sets of four variables for printing. Selecting *Mix and Match* allows you to pick any four of the available variables for printing. You must print four variables at a time in either mode. Depending on the program, you may be able to select other ancillary parameters such as the number of decimal places or the frequency of data to be printed.

Plotting Data (*Plot Button*)

The *Plot* button on the *Home* screen brings up the *PlotType* screen (see Figure A-2) which is the same in all programs. Variables in these programs can be plotted in one of several formats, three cartesian (see below) and one polar (see below). This screen allows a choice among these four “flavors” of plots shown as plot-style icons. The first icon (upper left) provides four functions plotted on cartesian axes in four separate windows. The second icon (upper right) plots two functions on cartesian axes in separate windows. The third icon (lower left) allows one to four functions to be plotted on common cartesian axes in a single window. This choice is of value to show a single function full screen or to overlay multiple functions. (Be advised, however, that multiple functions will scale to the largest function of the set, so if there are large differences in magnitude between the members of the set, it may be difficult to see and interpret the smaller ones.) The fourth icon (lower right) provides a polar plot of one selected function. You may select any of these four plot styles by clicking on its icon or on the *Select* button above it and then clicking *Done*. Double clicking on a plot icon will bring up the next screen immediately.

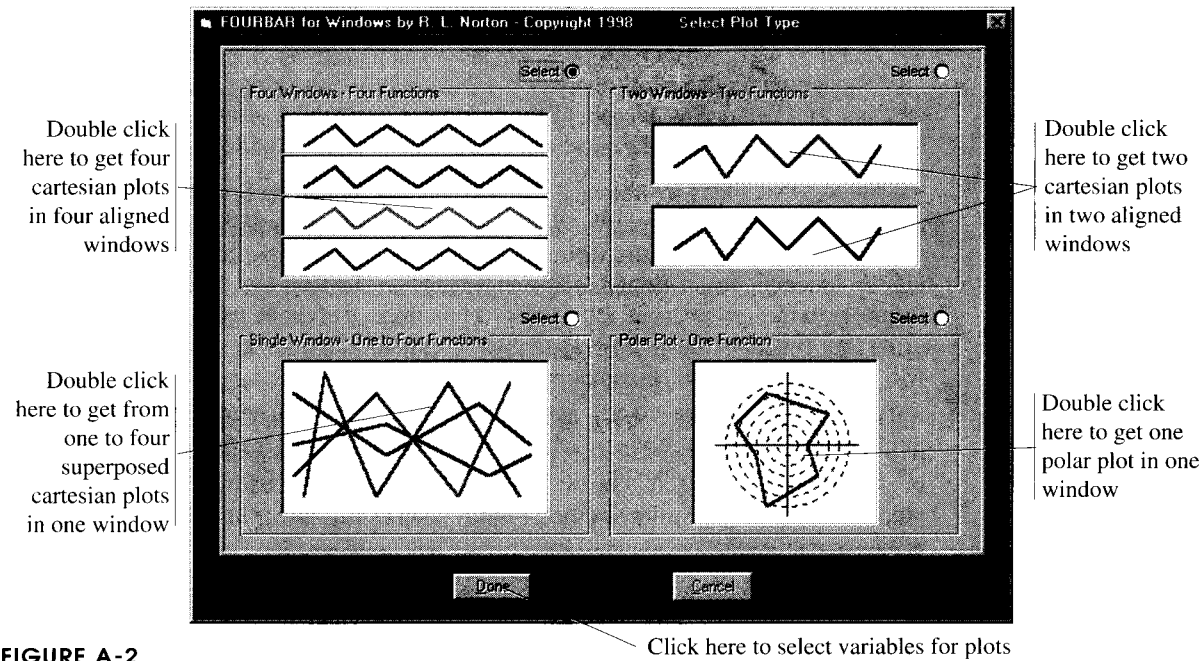


FIGURE A-2

The *PlotType* screen is common to all programs that allow plotting

CARTESIAN PLOTS depict a dependent variable versus an independent variable on cartesian (x, y) axes. In these programs, the independent variable shown on the x axis may be either time or angle, depending on the calculation choice made in the particular program. The variable for the y axis is selected from the plot menu. Angular velocities and torques are vectors but are directed along the z axis in a two-dimensional system. So their magnitudes can be plotted on cartesian axes and compared because their directions are constant, known, and the same.

POLAR PLOTS Plots of linear velocities, linear accelerations, and forces require a different treatment than the cartesian plots used for the angular vector parameters. Their directions are not the same and vary with time or input angle. One way to represent these linear vectors is to make two cartesian plots, one for magnitude and one for angle of the vector at each time or angle step. Alternatively, the x and y components of the vector at each time or angle step can be presented as a pair of cartesian plots. Either of these approaches requires two plots per vector and has the disadvantage of being difficult to interpret. A better method for vectors that act on a moving point (such as a force on a moving pin) can be to make a polar plot with respect to a local, nonrotating axis coordinate system (LNCS) attached at the moving point. This local, nonrotating x, y axis system translates with the point as it moves but remains always parallel to the global axis system X, Y . By plotting the vectors on this moving axis system we can see both their magnitude and direction at each time or angle step, since we are attaching the roots of all the vectors to the moving point at which they act.

In some of the programs, polar plots can be paused between the plotting of each vector. Without a pause, the plot can occur too quickly for the eye to detect the order in which they are drawn. When a mouse click is required between the drawing of each vector, their order is easily seen. With each pause, the current value of the independent variable (time or angle) as well as the magnitude and angle of the vector are displayed.

The programs also allow alternate presentations of polar plots, showing just the vectors, just the envelope of the path of the vector tips, or both. A plot that connects the tips of the vectors with a line (its envelope) is sometimes called a **hodograph**.

SELECTING PLOT VARIABLES Choosing anyone of the four plot types from the *Plot Type* screen brings up a *Plot Select* screen which is essentially the same in all programs. (See Figure A-3.) As with the *Print Select* screen, two arrangements for selecting the functions to be plotted are provided, *Preset Formats* and *Mix and Match*. The former provides preselected collections of functions, and the latter allows you to select up to four functions from those available on the pull-down menus. In some cases you will also have to select the component of the function desired, i.e., x , y , *mag*, or *angle*.

PLOT ALIGNMENT Some of the *Plot Select* screens offer a choice of two further plot style variants labeled *Aligned* and *Annotated*. The aligned style places multiple plots in exact phase relationship, one above the other. The annotated style does not align the plots but allows more variety in their display such as fills and grids. The data displayed is the same in each case.

COORDINATE SYSTEMS For particular variables in some programs, a choice of coordinate system is provided for display of vector information in plots. The *Coordinate System* panel on the *Plot Select* screen will become active when one of these variables is selected. Then either the *Global* or *Local* button can be clicked. (It defaults to *Global*.)

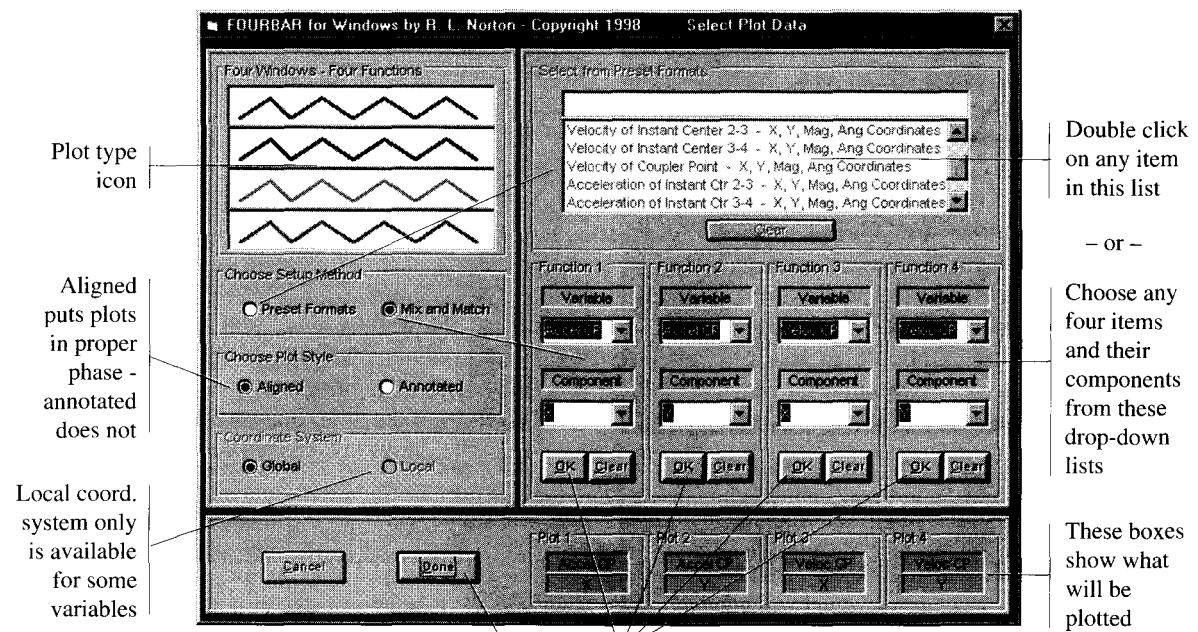
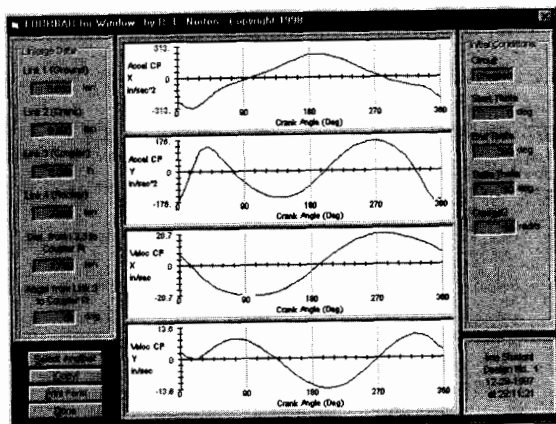


FIGURE A-3 Click here to see the plots — Click here to select a variable for plot
The *PlotSelect* screens are common to all programs. This shows one of four styles of *PlotSelect* screens

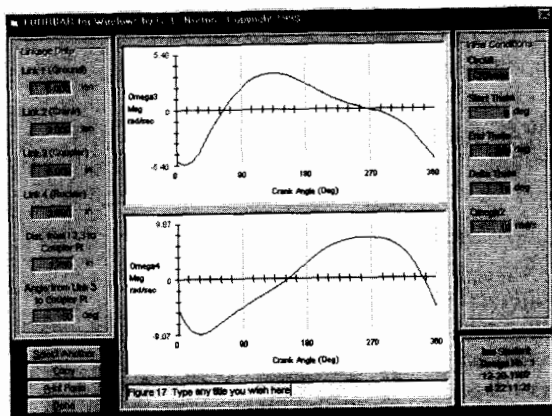
GLOBAL COORDINATES The *Global* choice in the *Coordinate System* panel refers all angles to the *XY* axes of Figure A-6 (p. 737). For polar plots the vectors shown with the *Global* choice actually are drawn in a local, nonrotating coordinate system (LNCS) that remains parallel to the global system such as x_1, Y_1 at point *A* and x_2, Y_2 at point *B* in Figure A-I. The LNCS X_2, Y_2 at point *B* behaves in the same way as the LNCS x_1, Y_1 at point *A*; that is, it travels with point *B* but remains parallel to the world coordinate system X, Y at all times.

LOCAL COORDINATES The coordinate system x', y' also travels with point *B* as its origin, but is embedded in link 4 and rotates with that link, continuously changing its orientation with respect to the global coordinate system X, Y making it an LRCS. Each link has such an LRCS but not all are shown in the figure. The *Local* choice in the *Coordinate System* panel uses the LRCS for each link to allow the plotting and printing of the tangential and radial components of acceleration or force on a link. This is of value if, for example, a bending stress analysis of the link is wanted. The dynamic force components perpendicular to the link due to the product of the link mass and tangential acceleration will create a bending moment in the link. The radial component will create tension or compression.

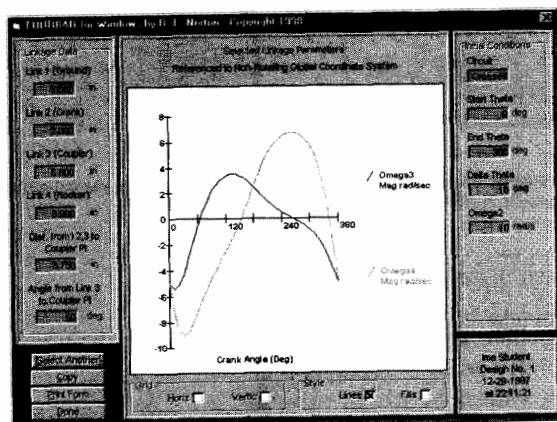
PLOTTING Once your selections are made and are shown in the cyan boxes at the lower right of the *Plot Select* screen, the *Done* button will become available. Clicking it will bring up the plots that you selected. Figure A-4 shows examples of the four plot types available. From this *Plot* screen you may copy to the clipboard for pasting into another application or dump the *Plot* screen to a printer. The *Select Another* button returns you to the previous *Plot Select* screen. *Done* returns you to the *Home* screen.



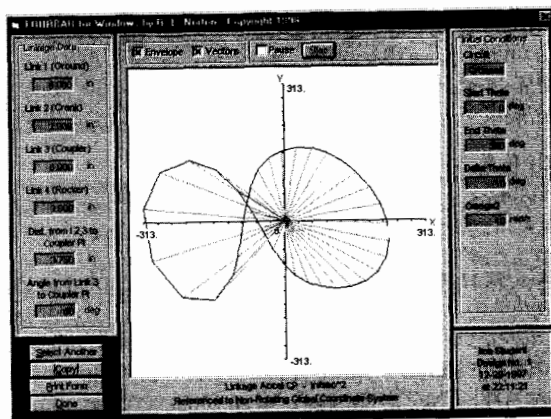
(a) Four aligned plots in separate windows



(b) Two aligned plots in separate windows



(c) One to four plots superposed in one window



(d) Single polar plot

FIGURE A-4

The four styles of plots available in all programs. Sidebar information is different in each program

The About Menu (All Programs)

The *About* pull-down menu on the *Home* screen will display a splash screen containing information on the edition and revision of your copy of the program. The *Disclaimer* and *Registration* form can also be accessed from this menu.

Exiting a Program (All Programs)

Choosing either the *Quit* button or *Quit* on the *File* pull-down menu on the *Home* screen will exit the program. If the current data has not been saved since it was last changed, it will prompt you to save the model using an appropriate suffix. In all cases, it will ask you to confirm that you want to quit. If you choose yes, the program will terminate and any unsaved data will be gone at that point.

Support (All Programs)

Please notify the author of any bugs via [email to rlnorton@wpi.edu](mailto:rlnorton@wpi.edu).

A.3 PROGRAM FOURBAR

FOURBAR for Windows is a linkage design and analysis program intended for use by students, engineers and other professionals who are knowledgeable in or are learning the art and science of linkage design. It is assumed that the user knows how to determine whether a linkage design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any linkage design, but cannot substitute for the engineering judgment of the user. The linkage theory and mathematics on which this program is based are documented in Chapters 4 to 7, 10, and 11 of this textbook. Please consult them for explanations of the theory and mathematics involved.

The *FOURBAR* Home Screen

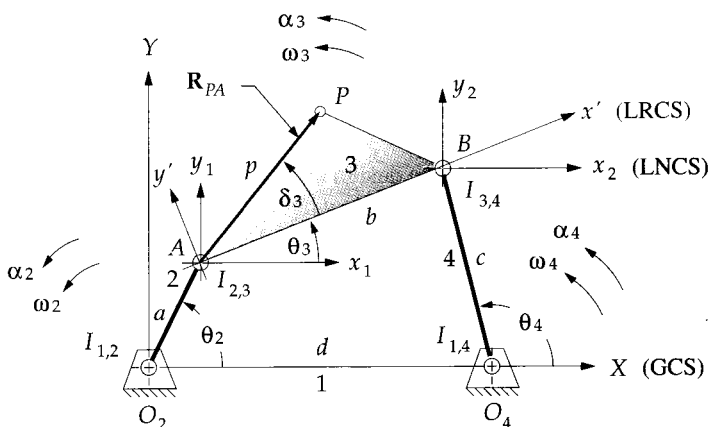
Initially, only the *Input* and *Quit* buttons are active on the *Home* screen. Typically, you will start a linkage design with the *Input* button, but for a quick look at a linkage as drawn by the program, one of the examples under the *Example* pull-down menu can be selected and it will draw a linkage. If you activate one of these examples, when you return from the *Animate* screen you will find all the other buttons on the *Home* screen to be active. We will address each of these buttons in due course below.

Input Data (*FOURBAR* Input Screen)

Figure A-5 defines the input parameters, link numbering, and the axis system used in program *FOURBAR*. The link lengths needed are ground link 1, input link 2, coupler link 3, and output link 4, defined by their pin-to-pin distances and labeled a , b , c , d in the figure. The X axis is constrained to lie along link 1, defined by the instant centers h_2 and h_4 which are also labeled, respectively, 02 and 04 in the figure. Instant center h_1 , the driver crank pivot, is the origin of the global coordinate system.

It might seem overly restrictive to force the X axis to lie on link 1 in this "aligned system." Many linkages will have their ground link at some angle other than zero. However, reorienting the linkage after designing and analyzing it merely involves rotating the piece of paper on which it is drawn to the desired final angle of the ground link. More formally, it means a rotation of the coordinate system through the negative of the angle of the ground link. In effect, the actual final angle of the ground link must be subtracted from all angles of links and vectors calculated in the aligned axis system.

In addition to the link lengths, you must supply the location of one coupler point on link 3 to find that point's coupler curve positions, velocities, and accelerations. This point is located by a position vector rooted at h_3 (point A) and directed to the coupler point P of interest which can be anywhere on link 3. The program requires that you input the polar coordinates of this vector which are labeled p and 03 in Figure A-5. The program asks for the distance from h_3 to the coupler point, which is p , and the angle the coupler point

**FIGURE A-5**

Linkage data for program FOURBAR. (Open the file F_A-05.5br in FOURBAR to see this linkage)

makes with link 3 which is α_3 . Note that angle α_3 is not referenced to either the global coordinate system (GCS) X, Y or to the local nonrotating coordinate system (LNCS) x, y at point A. Rather, it is referenced to the line AB which is the pin-to-pin edge of link 3 (LRCS). Angle α_3 is a property of link 3 and is embedded in it. The angle which locates vector R_{PA} in the x, y coordinate system is the sum of angle α_3 and angle δ_3 . This addition is done in the program, after δ_3 is calculated for each position of the input crank. Also see Section 4.5 (p. 152). The coordinate system, dimensions, angles, and nomenclature in Figure A-5 are consistent with those of Figure 4-6 (p. 154) which were used in the derivation of the equations used in program FOURBAR.

Calculation (FOURBARFIVEBAR\SIXBAR and SLIDER Input Screens)

Basic data for a linkage design is defined on the *Input* screen shown in Figure A-6, which is activated by selecting the *Input* button on the *Home* screen. When you open this screen for the first time, it will have default data for all parameters. The linkage geometry is defined in the *Linkage Data* panel on the left side of the screen. You may change these by typing over the data in the yellow text boxes.

The open or crossed circuit of the linkage is selected in a panel at the lower left of the screen in Figure A-6. Select the type of calculation desired, one of *Angle Steps*, *Time Steps*, or *One Position* from the *Calculation Mode* in the upper-right panel. The start, finish, and delta step information is different for each of these calculation methods, and the input text box labels in the *Initial Conditions* panel will change based on your choice. Type the desired initial, delta, and final conditions as desired.

ONE POSITION will calculate position, velocity, and acceleration for anyone specified input position θ_2 , input angular velocity ω_2 , and angular acceleration α_2 .

ANGLESTEPS assumes that the angular acceleration α_2 of input link 2 is zero, making ω_2 constant. The values of initial and final crank angle θ_2 , angle step $\Delta\theta_2$, and the constant input crank velocity ω_2 are requested. The program will compute all linkage pa-

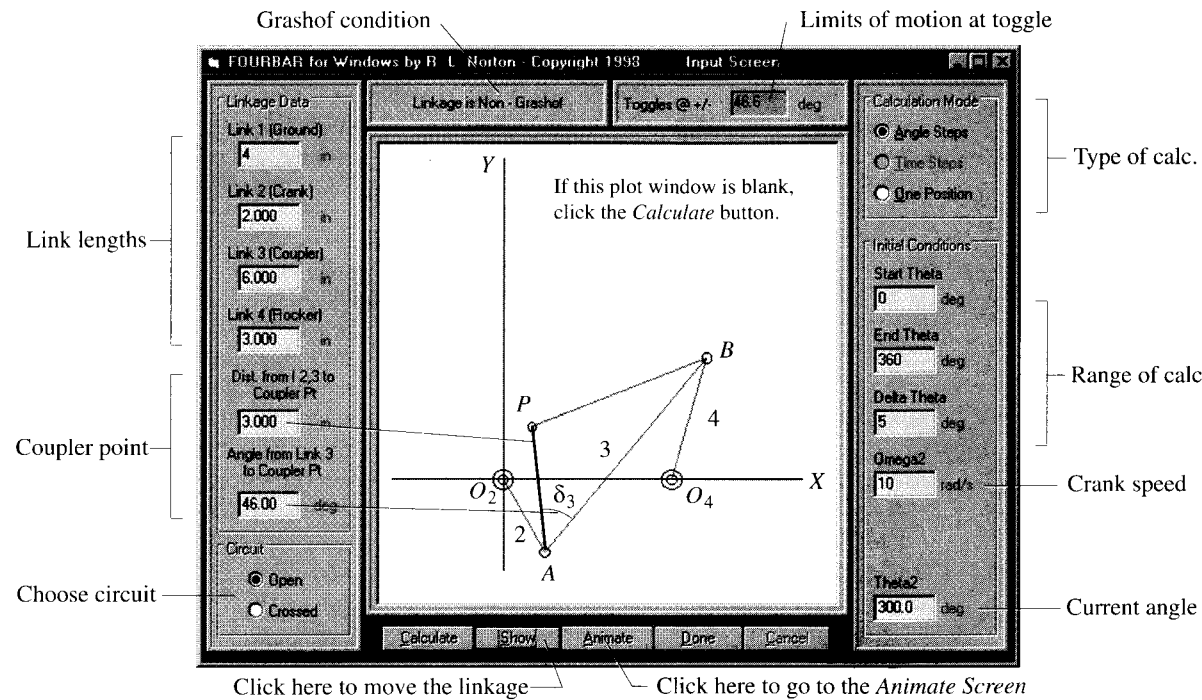


FIGURE A-6
Input Data screen for program FOURBAR. Corresponding screens in FIVEBAR, SIXBAR, and SLIDER are similar

rameters for each angle step. This is a steady-state analysis and is suitable for either Grashof or non-Grashof linkages provided that the total linkage excursion is limited in the latter case.

TIME STEPS requires input of a start time, finish time, and a time step, all in seconds. The value for ω_2 (which must be either a constant or zero) and the initial position θ_2 and initial velocity $\dot{\theta}_2$ of link 2 at time zero must also be supplied. The program will then calculate all linkage parameters for each time step by applying the specified acceleration, which of course will change the angular velocity of the driver link with time. This is a transient analysis. The linkage will make as many revolutions of the driver crank as is necessary to run for the specified time. This choice is more appropriate for Grashof linkages, unless very short time durations are specified, as a non-Grashof linkage will quickly reach its toggle positions.

Note that a combination of successive **Time Step**, **Crank Angle**, and **Time Step** analyses can be used to simulate the start-up, steady-state, and deceleration phases, respectively, of a system for a complete analysis.

CALCULATE The *Calculate* button will compute all data for your linkage and show it in an arbitrary position in the linkage window on the *Input* screen. If at any time the white linkage window is blank, the *Calculate* button will bring back the image. The *Show* button will move the linkage through its range in "giant steps."

After you have calculated the linkage, the *Animate* and *Done* buttons on the *Input* screen will become available. The *Animate* button takes you to the *Animate* screen where you can run the linkage through any range of motion to observe its behavior. You can also change any of the linkage parameters on the *Animate* screen and then recalculate the results there with the *Recalc* button. The *Done* button on either the *Input* or *Animate* screen returns you to the *Home* screen. The *Plot* and *Print* buttons will now be available as well as the *Animate* button which will send you to the *Animate* screen.

CALCULATION ERRORS If a position is encountered which cannot be reached by the links (in either the angle step or time step calculations), the mathematical result will be an attempt to take the square root of a negative number. The program will then show a dialog box with the message *Links do not connect for Theta2=xx* and present three choices: **Abort**, **Retry**, or **Ignore**. **Abort** will terminate the calculation at this step and return you to the *Input* screen. **Retry** will set the calculated parameters to zero at the current position and attempt to continue the computation at the next step, reporting successive problems as they occur. **Ignore** will continue the calculation for the entire excursion, setting the calculated parameters to zero at any subsequent positions with problems but will not present any further error messages. If a linkage is non-Grashof and you request calculation for angles that it cannot reach, then you will trip this error sequence. Choosing **Ignore** will force the calculation to completion, and you can then observe the possible motions of the linkage in the linkage window of the *Input* screen with the *Show* button.

GRASHOF CONDITION Once the calculation is done, the linkage's Grashof condition is displayed in a panel at the top left of the screen. If the linkage is non-Grashof, the angles at which it reaches toggle positions are displayed in a second panel at top right. This information can be used to reset the initial conditions to avoid tripping the "links cannot connect" error.

Animation (FOURBAR)

The *Animate* button on the *Home* screen brings up the *Animate* screen as shown in Figure A-7. Its *Run* button activates the linkage and runs it through the range of motion defined in its most recent calculation. The Grashof condition is reported at the upper right corner of the screen. The number of cycles for the animation can be typed in the *Cycles* box at the lower right of the screen.

A time delay (defaulted to 0) can be set with a drop-down menu below the *Cycles* box. Any number of cycles not on its list can be typed in the box. This time-delay feature is provided to accommodate variations in speed among computers. If your computer is very fast, the animation may occur too rapidly to be seen. If so, selecting larger positive numbers for the time delay will slow the animation. For slow computers, negative numbers will speed the animation to a degree but can make it jerky. The *Step* button moves the linkage one increment of the independent variable at a time.

Text boxes in the *Linkage Data* panel allow the linkage geometry to be changed without returning to the *Input* screen. The initial conditions can be redefined in the panel on the left of this screen. The *Open-Crossed* selection can be switched, but the *Calculation Type* can only be changed on the *Input* screen. After any such change, the linkage must be recalculated with the *Recalc* button and then rerun.

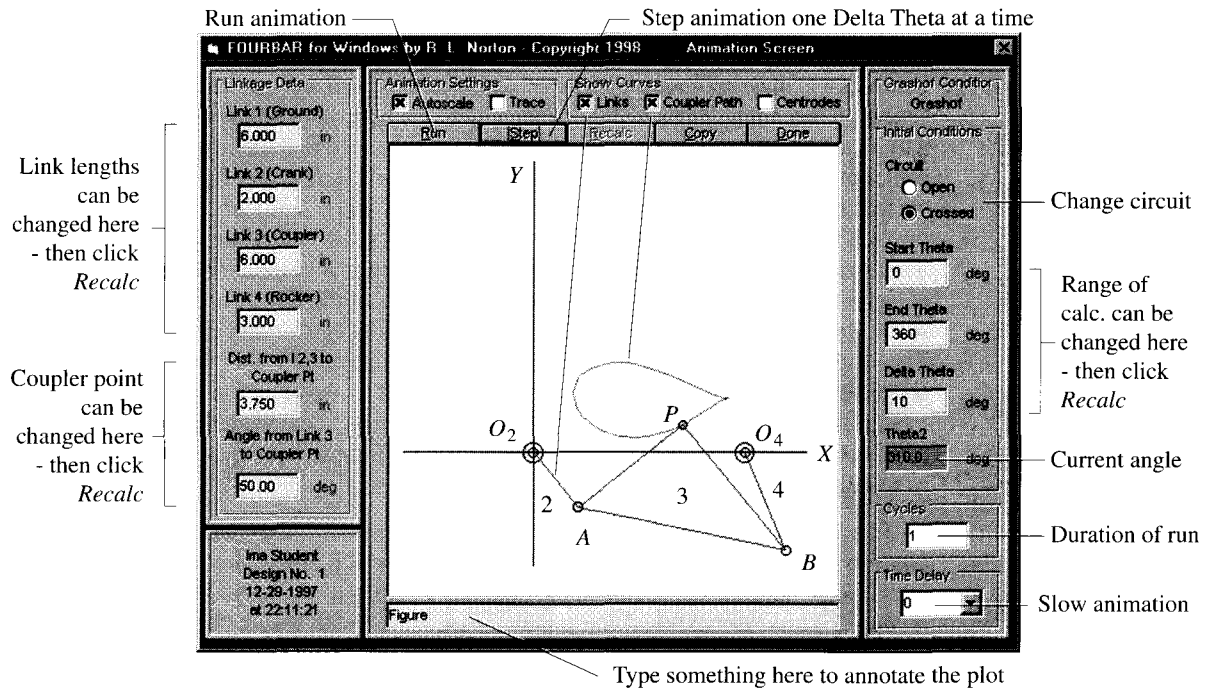


FIGURE A-7

FOURBAR Animation Screen. Programs FIVEBAR, SIXBAR, and SLIDER have similar animation screens

Two panels at the top center of the *Animate* screen provide switches to change the animation display. In the *Show Curves* panel, displays of *Links*, *Coupler Path*, and *Centroids* can be turned on and off in subsequent animations.

CENTRODES Only the FOURBAR program calculates and draws the fixed and moving centrodes (the loci of the instant centers as defined in Section 6.5 (p. 263) and shown in Figure 6-15a, p. 266). Different colors are used to distinguish the fixed from the moving centrode. The centrodes are drawn with their point of common tangency located at the first position calculated. Thus, you can orient them anywhere by your choice of start angle for the calculation.

AUTOSCALE can be turned on or off in the *Animation Settings* panel. The linkage animation plot is normally autoscaled to fit the screen based on the size of the linkage and its coupler curves (but not of the centrodes as they can go to infinity). You may want to turn off autoscaling when you wish to print two plots of different linkages at the same scale for later manual superposition. Turning off auto scaling will retain the most recent scale factor used. When on, it will rescale each plot to fit the screen.

TRACE Turning *Trace* on keeps all positions of the linkage visible on the screen so that the pattern of motion can be seen. Turning *Trace* off erases all prior positions, showing only the current position as it cycles the linkage through all positions giving a dynamic view of linkage behavior.

Dynamics (FOURBAR, FIVEBAR, SIXBAR, and SLIDER Dynamics Screen)

The *Dynamics* button on the *Home* screen brings up a screen that allows input of data on link masses, mass moments of inertia, *CG* locations, and any external forces or torques acting on the links. The location of the *CG* of each moving link is defined in the same way as the coupler point, by a position vector whose root is at the low-numbered instant center of each link. That is, for link 2 it is $I_{1,2}$; for link 3, $I_{2,3}$; and for link 4, $I_{1,4}$. These vectors for a fourbar linkage are shown in Figure A-2a (p. 731) labeled R_{CGi} where i is the link number. Note that the angle of each *CG* vector is measured with respect to a *local rotating coordinate system* (LRCS) whose origin is at the aforementioned instant center and whose x axis lies along the line of centers of the link. For example, in Figure A-8a, link 2's *CG* vector is 3 in at 30° , link 3's is 9 in at 45° , and link 4's is 5 in at 0° . The program will automatically create the necessary position vectors R_{12} , R_{32} , R_{23} , R_{43} , R_{34} , and R_{14} needed for the dynamic force equations as shown in the free-body diagrams in Figure A-8b. These position vectors are, of course, recalculated in the *nonrotating local coordinate systems* (LNCS) at the links' *CGs* for each new position of the linkage as the link angles change.

The masses and mass moments of inertia with respect to the *CGs* of the moving links are also required. Any external forces or torques which are applied to links 3 or 4 are typed in the appropriate boxes on the *Dynamics* screen as shown in Figure A-9. The direction angle of any external force must be measured with respect to the *global coordinate system* (GCS). The program will assume that this angle remains constant for all positions of the linkage analyzed. You must also supply the magnitude and direction of the position vector R_P which locates any point on the force vector F_P . This R_P vector is measured in the *rotating, local coordinate system* (LRCS) embedded in the link, as were the *CGs* of the links. R_P is *not* measured in the global system. The program takes care of the resolution of these R_P vectors, for each position of the linkage, into coordinates in the nonrotating local coordinate system at the *CG*. Note that if you wish to account for the gravitational force on a heavy link, you may do so by applying that weight as an external force acting through the link's *CG* at 270° in the global system ($R_P = 0$).

For other linkages such as the fivebar, sixbar, and fourbar slider-crank, the dynamic data input is similar. The only difference is the number of links for which mass property data and possible external forces and torques must be supplied.

After solving (by clicking the *Solve* button), clicking the *Show Matrix* button will display the dynamics matrix for the linkage. The results of the dynamics calculations are automatically stored for later plotting and printing. The menus on the *Plot* and *Print* screens will expand to include forces and torques for all links.

Balancing (FOURBAR Only)*

The *Balance* button on the FOURBAR *Home* screen brings up the *Balance* screen shown in Figure A-10, which immediately displays the mass-radius products needed on links 2 and 4 to force balance it and reduce the shaking force to zero. If you place the total amounts of the calculated mass-radius products on the rotating links, the shaking force will become zero and the shaking torque will increase. A partial balance condition can be specified by reducing the balance masses and accepting some nonzero shaking force in return for a smaller increase in torque.

* Program ENGINE also allows balancing but its *Balance Screen* is completely different than in FOURBAR and so will be discussed separately in the section on the ENGINE program.

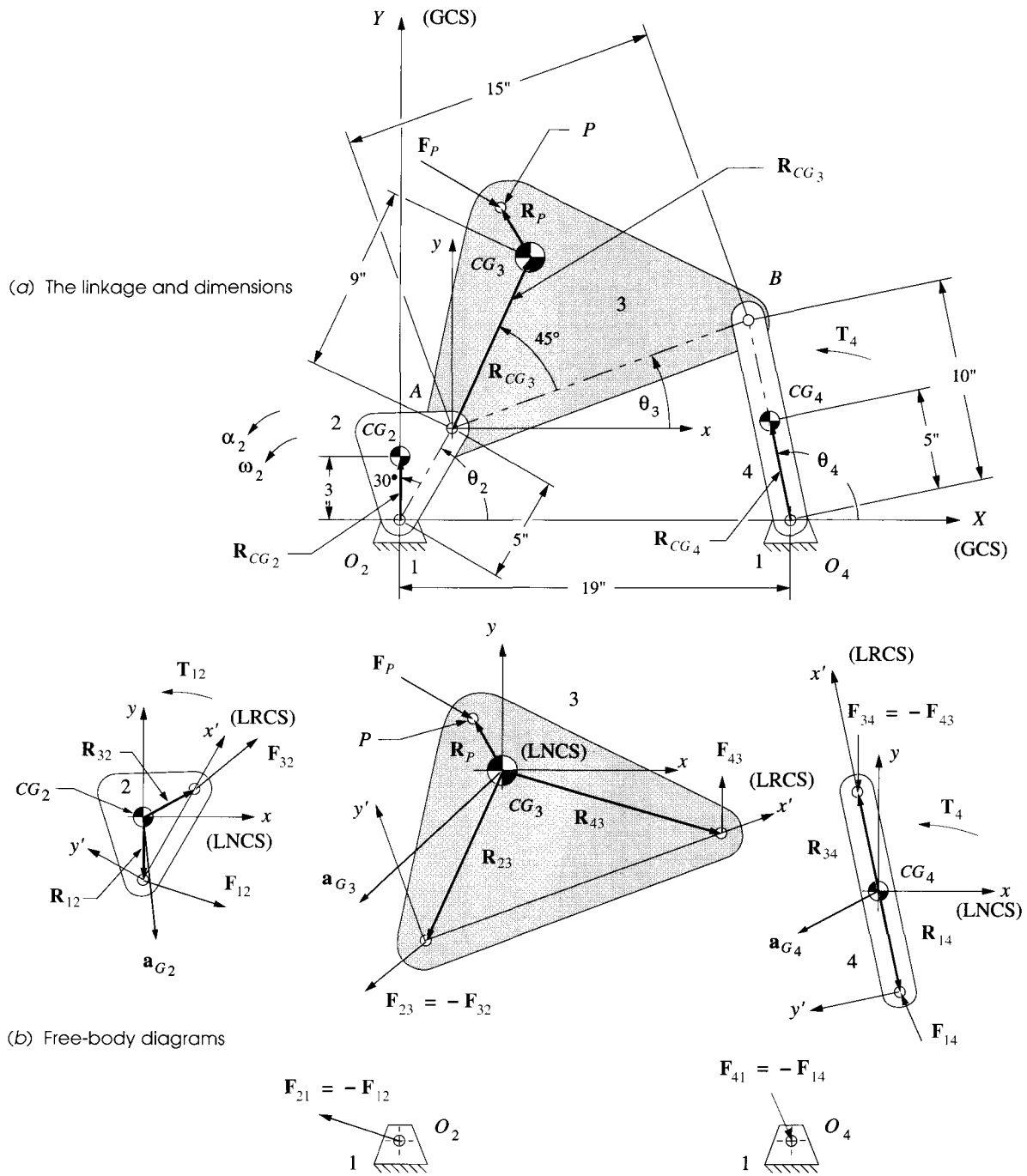


FIGURE A-8
Definition of data needed for dynamics calculations in Program FOURBAR

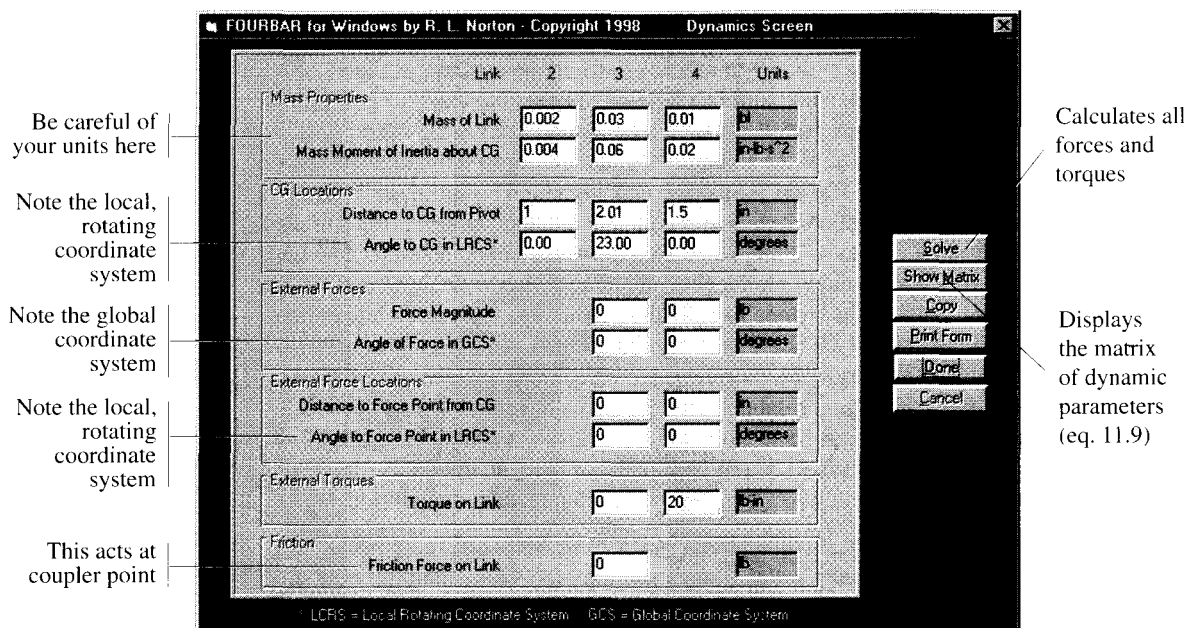


FIGURE A-9

Dynamic Data Input screen for program FOURBAR. Screens for programs FIVEBAR, SIXBAR, and SLIDER are similar

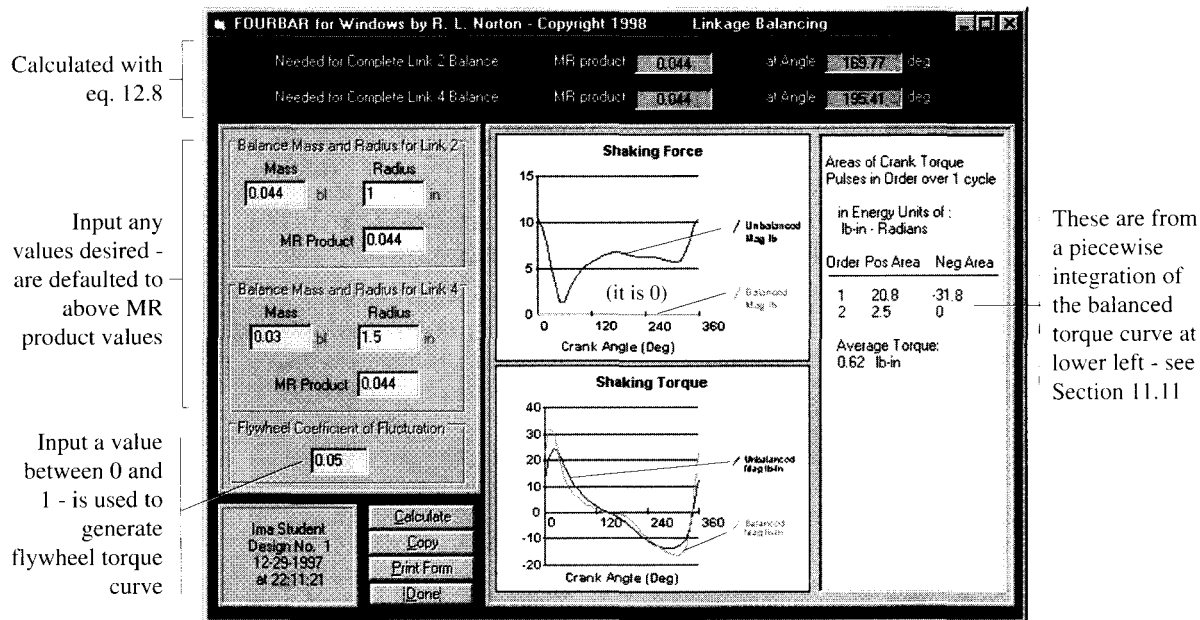


FIGURE A-10

Linkage Balancing screen in program FOURBAR only. Other programs do allow flywheel calculations

The **FOURBAR Balance** screen in Figure A-10 displays two mini-plots that superpose the shaking forces and shaking torques before and after balancing. Effects on the shaking force and torque from changes in the amount of balance mass-radius product placed on each link can be immediately seen in these plots. The energy in each pulse of the torque-time curve is also displayed in a sidebar on the right of this screen for use in a flywheel sizing calculation. See Section 11.11 (p. 548) for a discussion of the meaning and use of these data. The program calculates a smoothed torque function by multiplying the raw torque by the coefficient of fluctuation specified in the box at lower left.

Cognates (FOURBAR Only)

The *Cognates* pull-down menu allows switching among the three cognates which create the same coupler curve. Switching among them requires recalculation of all kinematic and dynamic parameters via the *Input*, *Dynamics*, and *Balance* buttons. The previously used mass property data is retained but can be changed easily by selecting the *Dynamics* button. The *Cayley Diagram* menu pick under *Cognates* displays that diagram of all three cognates. See Chapter 3. Whenever linkage data are changed on the *Input Screen* and recalculated, the program automatically calculates the dimensions of that linkage's two cognates. These can be switched to, calculated, and investigated at any time.

Synthesis (FOURBAR Only)

This pull-down menu allows selection of two- or three-position synthesis of a linkage, each with a choice of two methods. See Chapter 5 for a discussion of these methods and derivations of the equations used. When the linkage is synthesized, its link geometry is automatically put into the input sheet and recalculation is then required.

Other

See Section A.2, General Program Operation (p. 727) for information on *New*, *Open*, *Save*, *Save As*, *Plot*, *Print*, *Units*, and *Quit* functions.

A.4 PROGRAM FIVE BAR

FIVEBARfor Windows is a linkage design and analysis program intended for use by students, engineers, and other professionals who are knowledgeable in or are learning the art and science of linkage design. It is assumed that the user knows how to determine whether a linkage design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any geared fivebar linkage design, but cannot substitute for the engineering judgment of the user. The linkage theory and mathematics on which this program is based are documented in Chapters 4 to 7, and 11 of this textbook. Please consult them for explanations of the theory and mathematics involved.

The FIVEBAR Home Screen

Initially, only the *Input* and *Quit* buttons are active on the *Home* screen. Typically, you will start a linkage design with the *Input* button, but for a quick look at a linkage as drawn

by the program, one of the examples under the *Example* pull-down menu can be selected and it will draw a linkage. If you activate one of these examples, when you return from the *Animate* screen you will find all the other buttons on the *Home* screen to be active. We will address each of these buttons in due course below.

Input Data (The *Input* Screen)

Because this program deals with the geared fivebar mechanism (GFBM), it requires more input data than for the fourbar mechanism. Five link lengths must be supplied as shown in Figure A-11: driver crank (link 2), first coupler (link 3), second coupler (link 4), driven crank (link 5), and ground (link 1). Two other linkage parameters must be defined as input, namely, the gear ratio (λ) and the phase angle (ϕ). The phase angle is defined as the angle of link 5 when link 2 is at 0° as shown in Figure A-11. Note that the gear ratio as shown in the figure is a *negative ratio* because the external gears turn in opposite directions. The addition of an idler gear will create a positive gear ratio. It is also worth noting that the gear ratios defined in the *ZNH Atlas of Geared Fivebar Linkages* (Appendix E) are the reciprocal of the gear ratio in program FIVEBAR. So, when transferring data from this atlas to the program, the gear ratio must be reciprocated in order to get the same linkage as shown in the atlas. Otherwise the coupler curve will be a mirror image of the one in the atlas.

The coupler point P is defined in the same way as in the fourbar linkage. A coupler point can only be placed on link 3 in this program. If you want a coupler point on link 4, mirror your linkage and renumber the links to put the coupler point on link 3.

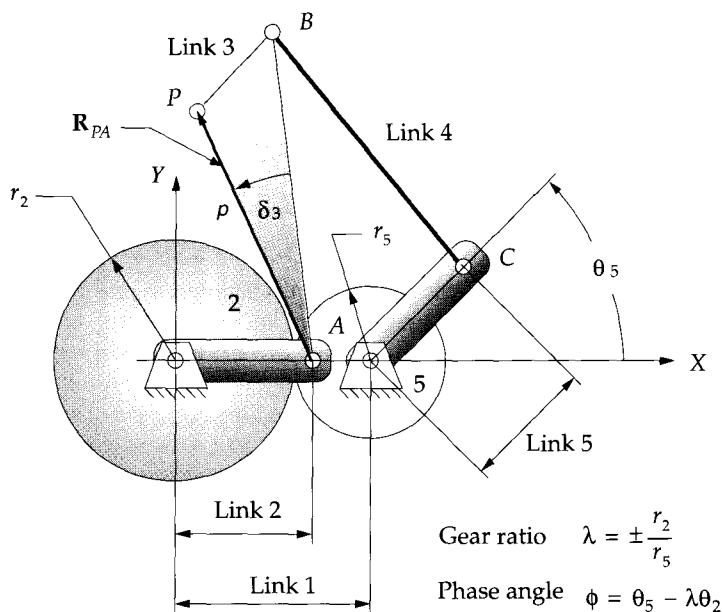


FIGURE A-11

Input data for program FIVEBAR. (Open the disk file F_A-11.5br in FIVEBAR to see this linkage)

Basic data for the fivebar linkage is defined on the *Input* screen which is similar to that for FOURBAR shown in Figure A-6 (p. 737). The *Input* screen is activated by selecting the *Input* button on the *Home* screen. When you open this screen for the first time, it will have default data for all link parameters. You may change any of these by typing over the data in the yellow text boxes.

Select the type of calculation desired in the upper right corner of the screen, one of *Angle Steps*, *Time Steps*, or *One Position*. The *Calculate* button will compute all data for your linkage and show it in an arbitrary position on the screen. If the white linkage display window is blank, the *Calculate* button will bring back the image. See the discussion of calculations for the FOURBAR program in Section A.3 (p. 735). They are similar in FIVEBAR.

After you have calculated the linkage, the *Animate* and *Done* buttons on the *Input* screen will become available. *Animate* takes you to the *Animate* screen where you can run the linkage through any range of motion to observe its behavior. You can also change any of the linkage parameters on the *Animate* screen and then recalculate the results with the *Recalc* button. The *Done* button on either the *Input* or *Animate* screen returns you to the *Home* screen. The *Plot* and *Print* buttons will now be available as well as the *Animate* button which returns you to the *Animate* screen.

Animation (FIVEBAR)

In program FIVEBAR, the *Animation* screen and its features are essentially similar to those of program FOURBAR. The only exception is the lack of a centrode selection in FIVEBAR. See the *Animation* discussion for FOURBAR in Section A-3 for more information.

Dynamics (FIVEBAR Dynamics Screen)

Input data for dynamics calculation in FIVEBAR are similar to that for program FOURBAR with the addition of one more link. However, linkage balancing is not available in FIVEBAR. See the discussion of dynamics calculations for program FOURBAR in Section A-3 for more information.

Other

See Section A.2, General Program Operation (p. 727) for information on *New*, *Open*, *Save*, *Save As*, *Plot*, *Print*, *Units*, and *Quit* functions.

A.5 PROGRAM SIXBAR

SIXBAR for Windows is a linkage design and analysis program intended for use by students, engineers, and other professionals who are knowledgeable in or are learning the art and science of linkage design. It is assumed that the user knows how to determine whether a linkage design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any linkage design but cannot substitute for the engineering judgment of the user. The linkage theory and mathematics on which this program is based are document-

ed in Chapters 4 to 10, and 11 of this textbook. Please consult them for explanations of the theory and mathematics involved.

The SIXBAR program is generally similar to the FOURBAR program. It will analyze the **Watt's II** and the **Stephenson's III** linkage isomers as defined in Figure 2-14 (p. 45). These are two of the five distinct sixbar isomers. The **Watt's II** mechanism is shown in Figure A-12 with the program's input parameters defined. The **Stephenson's III** mechanism is shown in Figure A-5 (p. 736) with its input parameters defined. Note that the program divides the sixbar into two stages of fourbar linkages. Stage 1 is the left half of the mechanism as shown in Figures A-12 and A-5. Stage 2 is the right half. The X axis of the global coordinate system is defined by instant centers $I_{1,2}$ and $I_{1,4}$ with its origin at $I_{1,2}$. The third fixed pivot $I_{1,6}$ can be anywhere in the plane. Its coordinates must be supplied as input.

The SIXBAR Home Screen

Initially, only the *Input* and *Quit* buttons are active on the *Home* screen. Typically, you will start a linkage design with the *Input* button, but for a quick look at a linkage as drawn by the program, one of the examples under the *Example* pull-down menu can be selected and it will draw a linkage. If you activate one of these examples, when you return from the *Animate* screen you will find all the other buttons on the *Home* screen to be active. We will address each of these buttons in due course below.

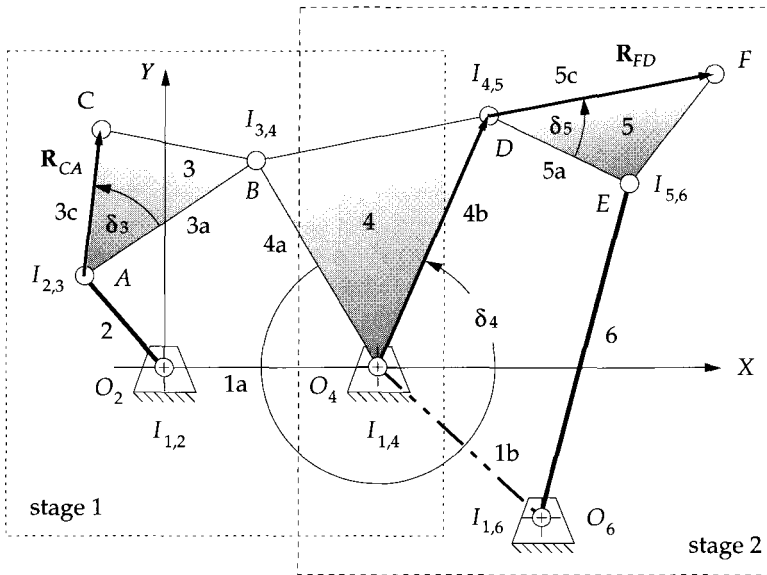
The *Home* screen's *Examples* pull-down menu includes both Watt's and Roberts' straight-line fourbar linkage stages driven by dyads (making them sixbars), a single-dwell sixbar linkage similar to that of Example 3-13 (p. 126) and Figure 3-31 (p. 127), and a double-dwell sixbar linkage that uses an alternate approach to that of Example 3-14 (p. 130) and Figure 3-32 (p. 129).

Input Data (SIXBARInput Screen)

Much of the basic data for the linkage design is defined on the *Input* screen which is activated by selecting the *Input* button on the *Home* screen. When you open this screen for the first time, it will have default data for all link parameters. You may change any of these by typing over the data in the yellow text boxes. A choice of Watt's or Stephenson's linkage must be made on the *Input* screen. The link information differs for the Watt's and Stephenson's linkages.

WATT'S II LINKAGE For the Watt's linkage (Figure A-12) the stage 1 data is: crank, first coupler, first rocker, and ground link segment from instant centers h_2 to h_4 . These correspond to links 2, 3a, 4a, and 1a, respectively, as labeled in the Figure A-12. The stage 2 data are: second crank, second coupler, second rocker, corresponding respectively to links 4b, 5a, and 6 in Figure A-12. The angle θ_4 that the second crank (4b) makes with the first rocker (4a) is also requested. Note that this angle obeys the right-hand rule as do all angles in these programs.

Two coupler points are allowed to be defined in this linkage, one on link 3 and one on link 5. The method of location is by polar coordinates of a position vector embedded in the link as was done for the fourbar and fivebar linkages. The first coupler point C is on link 3 and is defined in the same way as in FOURBAR. Program SrxBAR requires the

**FIGURE A-12**

Input data for program SIXBAR—a Watt's II linkage

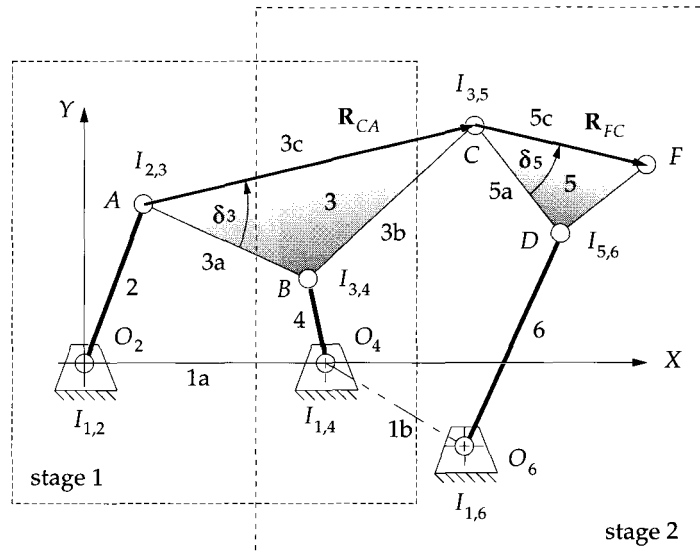
length (3c) of its position vector R_{CA} and the angle ϕ_3 which that vector makes with line 3a in Figure A-12. The second coupler point F is on link 5 and is defined with a position vector R_{FD} rooted at instant center $I_{4,5}$. The program requires the length (5c) of this position vector R_{FD} and the angle ϕ_5 which that vector makes with line 5a. The X and Y components of the location of the third fixed pivot $I_{1,6}$ are also needed. These are with respect to the global X, Y axis system whose origin is at $I_{1,2}$.

STEPHENSON'S III LINKAGE The stage I data for the Stephenson's linkage is similar to that of the Watt's linkage. The first stage's crank, first coupler, first rocker, and ground link segment from instant centers $I_{1,2}$ to $I_{1,4}$ correspond to links 2, 3a, 4, and 1a, respectively, in Figure A-13. The link lengths in stage 2 of the linkage are: second coupler and second rocker corresponding, respectively, to links 5a and 6 in Figure A-B.

Note that, unlike the Watt's linkage, there is no "second crank" in the Stephenson's linkage because the second stage is driven by the coupler (link 3) of the first stage. In this program link 5 is constrained to be connected to link 3 at link 3's defined coupler point C which then becomes instant center $I_{3,5}$. The data for this are requested in similar format to the Watt's linkage, namely, the length of the position vector R_{CA} (line 3c) and its angle ϕ_3 . The second coupler point location on link 5 is defined, as before, by position vector R_{FC} with length 5c and angle ϕ_5 .

The X and Y components of the location of the third fixed pivot $I_{1,6}$ are required. These are with respect to the global X, Y axis system.

Select the type of calculation desired in the upper right corner of the screen, one of *Angle Steps*, *Time Steps*, or *One Position*. The *Calculate* button will compute all data for your linkage and show it in an arbitrary position on the screen. If at any time the link-

**FIGURE A-13**

Input data for program SIXBAR—a Stephenson's III linkage

age display window is blank, the *Calculate* button will bring back the image. See the description of calculations for program FOURBAR in Section A.3 (p. 735). They are similar in SIXBAR.

After you have calculated the linkage, the *Animate* and *Done* buttons on the *Input* screen will become available. *Animate* takes you to the *Animate* screen where you can run the linkage through any range of motion to observe its behavior. You can also change any of the linkage parameters on the *Animate* screen and then recalculate the results with the *Recalc* button. The *Done* button on either the *Input* or *Animate* screen returns you to the *Home* screen. The *Plot* and *Print* buttons will now be available as well as the *Animate* button which returns you to the *Animate* screen.

Animation (SIXBAR)

The *Animation* screen and its features in SIXBAR are essentially similar to those of Program FOURBAR. The only exception is the lack of a *Centrode* selection in SIXBAR. See the *Animation* discussion for FOURBAR in Section A.3 for more information.

Dynamics (SIXBAR Dynamics Screen)

Input data for dynamics calculation in SIXBAR are similar to that for program FOURBAR with the addition of two links. See the discussion of dynamics calculations for program FOURBAR in Section A.3 for more information.

Other

See Section A.2, General Program Operation (p. 727) for information on *New*, *Open*, *Save*, *Save As*, *Plot*, *Print*, *Units*, and *Quit* functions.

A.6 PROGRAM SLIDER

SLIDERfor Windows is a linkage design and analysis program intended for use by students, engineers, and other professionals who are knowledgeable in or are learning the art and science of linkage design. It is assumed that the user knows how to determine whether a linkage design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any linkage design, but cannot substitute for the engineering judgment of the user. The linkage theory and mathematics on which this program is based are documented in Chapters 4 to 10, and 11 of this textbook. Please consult them for explanations of the theory and mathematics involved.

The Slider Home Screen

Initially, only the *Input* and *Quit* buttons are active on the *Home* screen. Typically, you will start a linkage design with the *Input* button, but for a quick look at a linkage as drawn by the program, one of the examples under the *Example* pull-down menu can be selected and it will draw a linkage. If you activate one of these examples, when you return from the *Animate* screen you will find all the other buttons on the *Home* screen to be active. We will address each of these buttons below.

Input Data (SLIDER Input Screen)

Basic data for the sixbar linkage is defined on the *Input* screen, which is activated by selecting the *Input* button on the *Home* screen. When you open this screen for the first time, it will have default data for all link parameters. You may change any of these by typing over the data in the yellow text boxes.

Figure A-14 defines the input parameters for the fourbar slider-crank linkage. The link lengths needed are input link 2 and coupler link 3, defined by their pin-to-pin distances and labeled a and b in the figure. The X axis lies along the line d , through instant center $h,2$ (point $O2$) and parallel to the direction of motion of slider 4. Instant center $h,2'$ the driver crank pivot, is the origin of the global coordinate system. The slider offset c is the perpendicular distance from the X axis to the sliding axis. Slider position d will be calculated for all positions of the linkage.

In addition to the link lengths, you must supply the location of one coupler point on link 3 to find that point's coupler curve positions, velocities, and accelerations. This point is located by a position vector rooted at $h,3$ (point A) and directed to the coupler point P of interest which can be anywhere on link 3. The program requires that you input the polar coordinates of this vector which are labeled p and $B3$ in Figure A-14. The program needs the distance from $h,3$ to the coupler point p and the angle $B3$ that the coupler point makes with link 3. Note that angle $B3$ is not referenced to either the global coordinate system

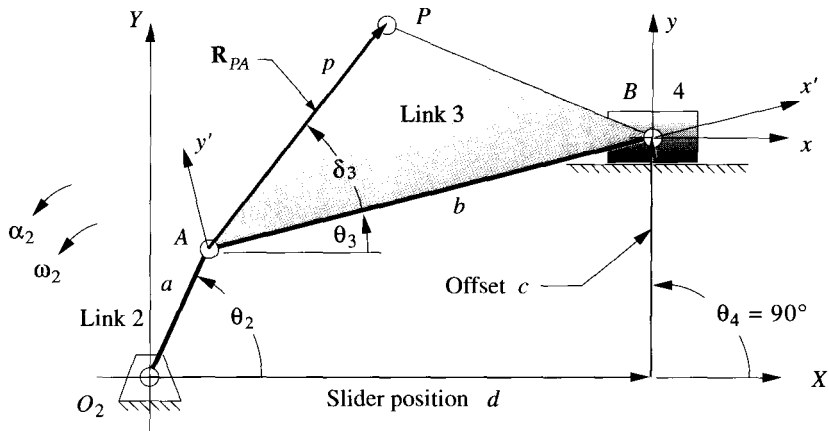


FIGURE A-14

Input data for Program SLIDER

X, Y or to the local nonrotating coordinate system x, y at point A. Rather, it is referenced to the line AB which is the pin-to-pin edge of link 3. Angle δ_3 is a property of link 3 and is embedded in it. The angle which locates vector R_{CA} in the x, y coordinate system is the sum of angle δ_3 and angle θ_3 . This addition is done in the program, after δ_3 is calculated for each position of the input crank.

The definition of δ_3 is different in program SLIDER than in Figure 4-9 (p. 160) and the derivations in Section 4.6 (p. 159). This was done to maintain consistency of input data among all the linkage programs in this package. Actually, within program SUDER, the angle shown in Figure A-14 that you input as δ_3 is converted using the law of cosines to the one shown in Figure 4-9 before the calculations are done with the equations from Section 4.6.

Calculation (SLIDER Input Screen)

See the description of calculations for program FOURBAR in Section A.3 (p. 735) for more information. They are similar in SUDER. Select the type of calculation desired in the upper right corner of the screen, one of *Angle Steps*, *Time Steps*, or *One Position*. The *Calculate* button will compute all data for your linkage and show it in an arbitrary position on the screen. If at any time the white linkage window is blank, the *Calculate* button will bring back the image.

After you have calculated the linkage, the *Animate* and *Done* buttons on the *Input* screen will become available. *Animate* takes you to the *Animate* screen where you can run the linkage through any range of motion to observe its behavior. You can also change any of the linkage parameters on the *Animate* screen and then recalculate the results with the *Recalc* button. The *Done* button on either the *Input* or *Animate* screen returns you to the *Home* screen. The *Plot* and *Print* buttons will now be available as well as the *Animate* button which returns you to the *Animate* screen.

Animation (SLIDER *Animation* Screen)

The *Animation* screen and its features in SLIDER are essentially similar to those of program FOURBAR. An exception is the lack of a *Centrode* selection in SLIDER. See the discussion of the *Animation* screen for FOURBAR in Section A.3 for more information.

Dynamics (SLIDER *Dynamics* Screen)

Input data for dynamics calculation in SLIDER are similar to that for program FOURBAR. See the discussion of dynamics calculations for Program FOURBAR in Section A.3 for more information.

Other

See Section A.2, General Program Operation (p. 727) for information on *New*, *Open*, *Save*, *Save As*, *Plot*, *Print*, *Units*, and *Quit* functions.

A.7 PROGRAM DYNACAM

DYNACAM *for Windows* is a cam design and analysis program intended for use by students, engineers, and other professionals who are knowledgeable in the art and science of cam design. It is assumed that the user knows how to determine whether a cam design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any cam design but cannot substitute for the engineering judgment of the user. The cam theory and mathematics on which this program is based are shown in Chapter 8 and 15 of this text. Please consult them for complete explanations of the theory and mathematics involved.

The DYNACAM *Home* Screen

Initially, only the *SVAJ* and *Quit* buttons are active on the *Home* screen. Typically, you will start a cam design with the *SVAJ* button, but for a quick look at a cam as drawn by the program, one of the examples under the *Example* pull-down menu can be selected and it will draw a cam profile. If you activate one of these examples, when you return from the *Cam Profile* screen you will find all the other buttons on the *Home* screen to be active. We will address each of these buttons in due course below.

Input Data (DYNACAM *Input* Screen)

Much of the basic data for the cam design is defined on the *Input* screen shown in Figure A-15, which is activated by selecting the *SVAJ* button on the *Home* screen. When you open this screen for the first time, it will be nearly blank, with only one segment's row visible. (Note that the built-in examples can also be accessed from this form at its upper right corner.) If you selected an example cam from the pull-down menu on the *Home* screen, making the *Input* screen nonblank, please now select the *Clear All* button on the *Input* screen to zero all the data and blank the screen, in order to better follow the presentation below. We will proceed with the explanation as if you were typing your data into an initially blank *Input* screen.

These can only be changed in the professional edition

Pull-down for examples

You can plot or print each segment's data to inspect results as you go

Click Calc for each segment

Seg	Beta	Start	End	Motion	Program	Position Start	Position End
1	50	0	50	Rise	MS - Mod Sine	0	1
2	40	50	90	Dwell	DW - Dwell	1	1
3	50	90	140	Fall	CF - Cycloid Full	1	0
4	40	140	180	Dwell	DW - Dwell	0	0
5	50	180	230	Poly	PO - Polynomial	0	1
6	40	230	270	Dwell	DW - Dwell	1	1
7	50	270	320	Fall	MT - Mod Trap	1	0
8	40	320	360	Dwell	DW - Dwell	0	0

FIGURE A-15

S V A J input screen for program DYNACAM

If you use the *Tab* button, it will lead you through the steps needed to input all data. On a blank *Input* screen, *Tab* first to the *Cam Omega* box in the upper left corner and type in the speed of the camshaft. *Tab* again (or mouse click if you prefer) to the *Number of Segments* box and type in any number desired between 1 and 16. That number of rows will immediately become visible on the *Input* screen. Note that some of this screen's choices such as *Osc* (oscillating arm follower) and *Delta Theta* are disabled in the student edition of the program.

Another *Tab* should put your cursor in the box for the *Beta* (segment duration angle) of segment 1. Type any desired angle (in degrees). Successive *Tabs* will take you to each *Beta* box to type in the desired angles. All *Betas* must, of course, sum to 360 degrees. If they do not, a warning will appear.

As you continue to *Tab* (or click your mouse in the appropriate box, if you prefer), you will arrive at the boxes for *Motion* selection. These boxes offer a pull-down selection of *Rise*, *Fall*, *Poly*, and *Dwell*. You may select from the pull-down menus with the mouse, or you can type the first letter of each word to select them. *Rise*, *fall*, and *dwell* have obvious meanings. The *Poly* choice indicates that you wish to create a customized polynomial function for that segment, and this will later cause a new screen to appear on which you will define the boundary conditions of your desired polynomial function.

The next set of choices that you will *Tab* or mouse click to are the *Program* pull-downs. These provide a menu of standard cam functions such as *Modified Trapezoid*, *Modified Sine*, and *Cycloid*. Also included are portions of functions such as the first and

second halves of cycloids and simple harmonics that can be used to assemble piecewise continuous functions for special situations. See Chapter 8 for additional information.

After you have selected the desired *Program* functions for each segment, you will *Tab* or *Click* to the *Position Start* and *End* boxes. *Start* in this context refers to the beginning displacement position for the follower in the particular segment, and *End* for its final position. You may begin at the "top" or "bottom" of the displacement "hill" as you wish, but be aware that the range of position values of the follower must be from a zero value to some positive value over the whole cam. In other words, you cannot include any anticipated base or prime circle radius in these position data. These position numbers represent the excursion of the so-called S diagram (displacement) of the cam and cannot include any prime circle information (which will be input later).

As each row's (segment's) input data are completed, the *Calc* button for that row will become enabled. Clicking on this button will cause that segment's *S*, *V*, *A*, and *J* data to be calculated and stored. After the *Calc* button has been clicked for any row, the *Plot* and *Print* buttons for that segment will become available. Clicking on these buttons will bring up a plot or a printed table of data for *S*, *V*, *A*, *J* data for that segment only. More detailed plots and printouts can be obtained later from the *Home* screen.

Polynomial Functions

If any of your segments specified a *Poly* motion, clicking the *Calc* button will bring up the *Boundary Condition* screen shown in Figure A-16. The cursor will be in the box for *Number of Conditions Requested*. Type the number of boundary conditions (BCs) de-

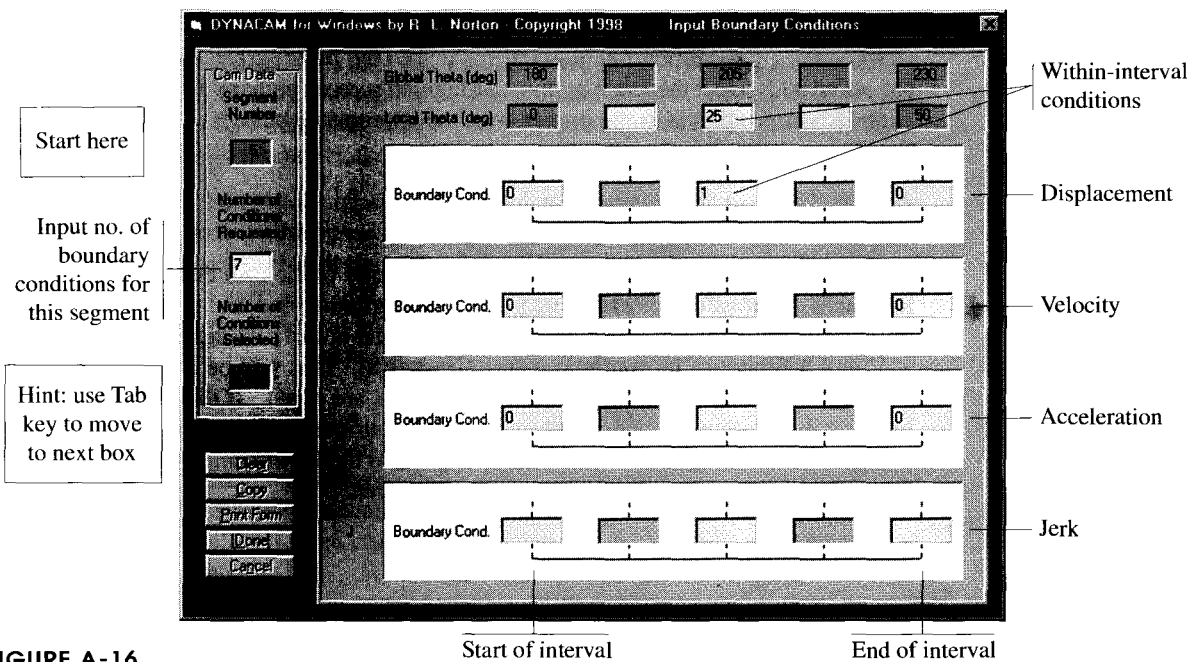


FIGURE A-16 Boundary Condition Input screen for polynomial functions in program DYNACAM—3-4-5-6 single-dwell function shown

sired, which must be between 2 and 20 inclusive. When you hit *Enter* or *Tab* or mouse click away from this box, the rest of the screen will activate, allowing you to type in the desired values of BCs. Note that the start and end values of position that you typed on the *Input* screen are already entered in their respective S boundary condition boxes at the beginning and end of the segment. Type your additional end of interval conditions on *V*, *A*, and *J* as desired. If you also need some BCs within the interval, click or tab to one of the boxes in the row labeled *Local Theta* at the top of the screen and type in the value of the angle at which you wish to provide a Be. That column will activate and you may type whatever additional BCs you need.

The box labeled *Number of Conditions Selected* monitors the BC count, and when it matches the *Number of Conditions Requested*, the *Done* button becomes available. Note that what you type in any (yellow) text box is not accepted until you hit *Enter* or move off that box with the *Tab* key or the mouse, allowing you to retype or erase with no effect until you leave the text box. (This is generally true throughout the program.)

Selecting the *Done* button from the BC screen calculates the coefficients of the polynomial by a Gauss-Jordan reduction method with partial pivoting. All computations are done in double precision for accuracy. If an inconsistent set of conditions is sent to the solver, an error message will appear. If the solution succeeds, it calculates $s v a j$ for the segment. When finished, it brings up a summary screen that shows the BCs you selected and also the coefficients of the polynomial equation that resulted. You may at this point want to print this screen to the printer or copy and paste it into another document for your records. You will only be able to reconstruct it later by again defining the BCs and recalculating the polynomial.

Back to the *Input* Screen

Completing a polynomial function returns you to the *Input* screen. When all segments are calculated, select the *Done* button on the *Input* screen (perhaps after copying it to the clipboard or printing it to the printer with the appropriate buttons). This will bring up the *Continuity Check* screen.

Continuity Check Screen

This screen provides a visual check on the continuity of the cam design at the segment interfaces. The values of each function at the beginning and end of each segment are grouped together for easy viewing. The fundamental law of cam design requires that the *S*, *V*, and *A* functions be continuous (see Chapter 8). This will be true if the boundary values for those functions shown grouped as pairs are equal. If this is not true, then a warning dialog box will appear when the *Done* button is clicked. This will not prevent you from proceeding but will suggest you return to the *Input Data* screen to correct the problem. Then it will return you to the *Home* screen.

Sizing the Cam

Once the *S V A J* functions have been defined to your satisfaction, it remains to size the cam and determine its pressure angles and radii of curvature. This is done from the *Size Cam* screen shown in Figure A-17, which is accessible from the button of that name on

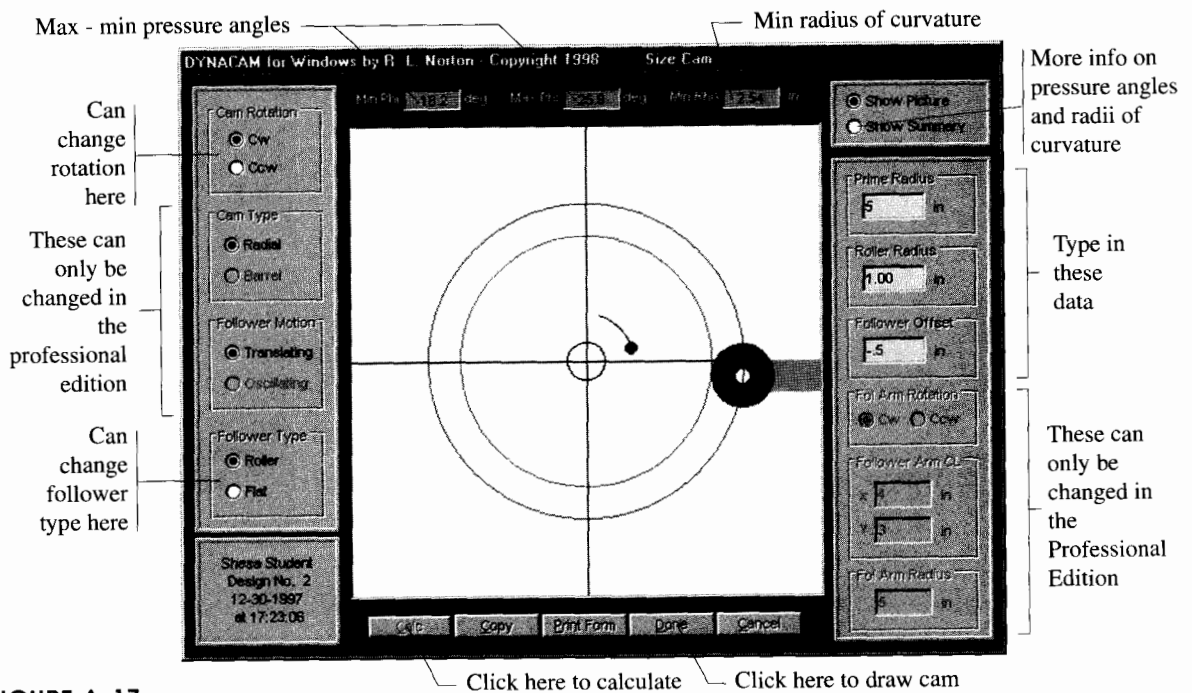


FIGURE A-17

Size Cam screen from DYNACAM

the *Home* screen. The *Size Cam* screen allows the cam rotation direction and follower type (flat or roller) to be set. The cam type (radial or barrel) and follower motion (translating or oscillating) are only available in the professional version. The prime circle radius and follower offset for a translating follower can be typed into their boxes. When a change is made to any of these parameters, the schematic image of cam and follower is updated.

Select the *Calc* button to compute the cam size parameters. The max and min pressure angles will appear in the boxes at the top of the screen. For more information, select the *Show Summary* button at the upper right. This will change the schematic image to a summary of pressure angle and radius of curvature information. When you hit the *Done* button, the cam profile will be drawn.

Drawing the Cam

The cam profile shown when leaving the *Size Cam* screen can be independently accessed at any time from the *Home* screen with the *Draw Cam* button. The *Draw Cam* screen shown in Figure A-18, allows the cam rotation direction, prime circle radius, roller radius, and offset (if available) to be changed. The cam size data will be recalculated and the new profile displayed.

In the cam profile drawing, a curved arrow indicates the direction of cam rotation. The initial position of a roller follower at cam angle $\theta = 0$ is shown as a filled circle with rectangular stem, and of a flat-faced follower as a filled rectangle. Any eccentricity

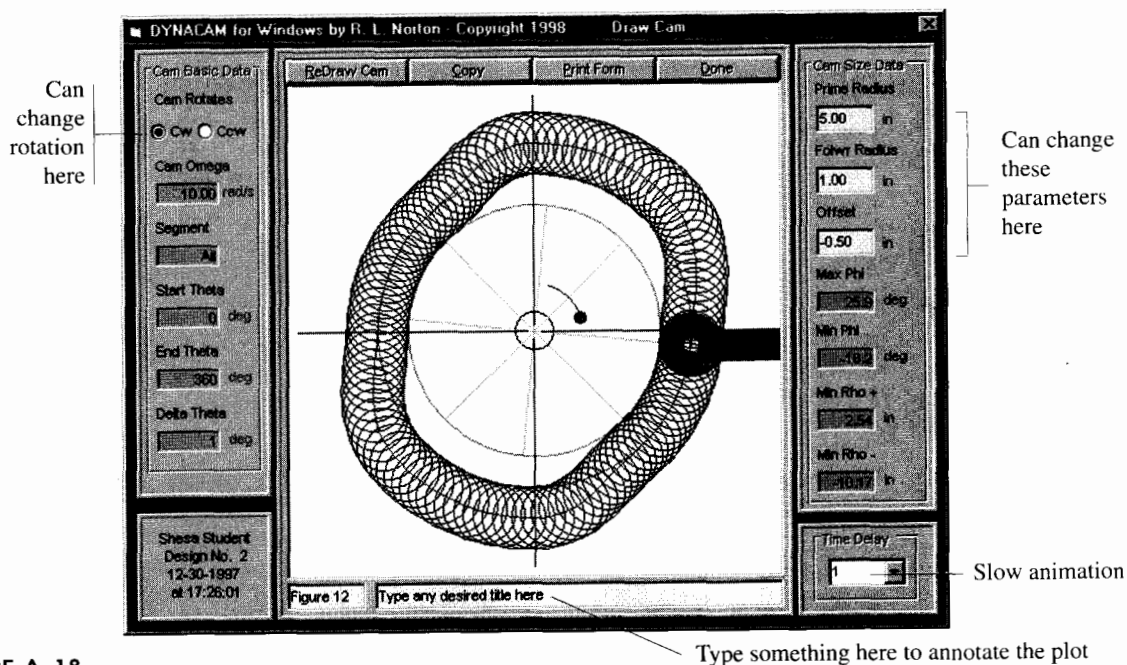


FIGURE A-18

Cam Profile screen in program DYNACAM

shows as a shift up or down of the filled follower with respect to the X axis through the cam center. The smallest filled circle on the cam centerline represents the camshaft. The smallest unfilled circle is the base circle. The prime circle may be somewhat obscured by the repeated drawings of the follower diameter, which sweep out the cam surface. The pitch curve is drawn along the locus of the roller follower centers. The radial lines that form pieces of pie within the base circle represent the segments of the cam. If the cam turns counterclockwise, the radial lines are numbered clockwise around the circumference and vice versa.

Follower Dynamics (DvNACAMOnly)

When the cam has been sized, the *Dynamics* button on the *Home* screen will become available. This button brings up the *Dynamics* screen shown in Figure A-19. Text boxes are provided for typing in values of the effective mass of the follower system, the effective spring constant, the spring preload, and a damping factor. By effective mass is meant the mass of the entire follower system as reflected back to the cam-follower roller centerline or cam contact point. Any link ratios between the cam-follower and any physical masses must be accounted for in calculating the effective mass. Likewise the effective spring in the system must be reflected back to the follower. The damping is defined by the damping ratio zeta, as defined for second-order vibrating systems. See Chapter 10 for further information.

The journal diameter and the coefficients of friction are used for calculating the friction torque on the shaft. The *Start New* or *Accumulate* switch allows you to either make

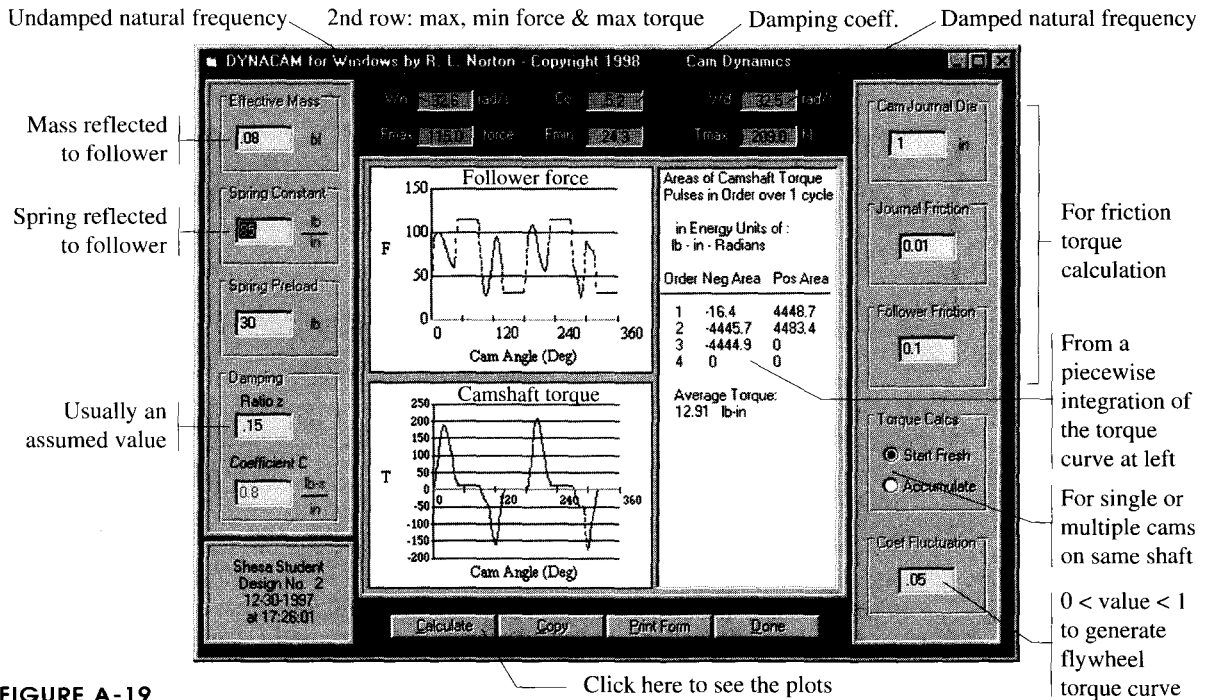


FIGURE A-19

Cam Dynamics screen in DYNACAM

a fresh torque calculation or accumulate the torques for several cams running on a common shaft. The energy information in the window can be used to calculate a flywheel needed for any coefficient of fluctuation chosen as described in Section 11.11 (p.548). The program calculates a smoothed torque function by multiplying the raw camshaft torque by the coefficient of fluctuation specified in the box at lower right of the screen.

Other

See Section A.2, General Program Operation (p. 727) for information on *New*, *Open*, *Save*, *Save As*, *Plot*, *Print*, *Units*, and *Quit* functions.

A.8 PROGRAM ENGINE

ENGINE for Windows is an internal combustion (Ic) engine design and analysis program intended for use by students, engineers, and other professionals who are knowledgeable in the art and science of engineering design. It is assumed that the user knows how to determine whether a design is good or bad and whether it is suitable for the application for which it is intended. The program will calculate the kinematic and dynamic data associated with any engine design, but cannot substitute for the engineering judgment of the user. The theory and mathematics on which this program is based are shown in Chapters 13 and 14 of this textbook. Please consult them for an explanation of the theory and mathematics involved.

The ENGINE Home Screen

Initially, only the *Input* and *Quit* buttons are active on the *Home* screen. Typically, you will start a design with the *Input* button, but before doing so, one of the examples under the *Example* pull-down menu can be selected. If you activate one of these examples, it will calculate the result and take you to the *Engine Data* screen (see Figure A-22). Click on the *Run* button to see the engine in operation. When you return to the *Home* screen you will find all the other buttons to be active. We will address each of these buttons below.

Single-Cylinder Engine (*Input One-Cylinder Data* Screen)

This program can be thought of as being used in two stages, each of which roughly corresponds to the topics in Chapters 13 and 14, respectively. That is, the *Input* and *Balance* buttons on the *Home* screen deal with single-cylinder engines (Chapter 13), and the *Assemble* button with multicylinder engines (Chapter 14). The *Flywheel* button can be used with either a single- or multicylinder design.

THE *INPUT* BUTTON on the *Home* screen brings up the *Input One-Cylinder Data* screen shown in Figure A-20. Default data are present in all entry boxes. Device type can be set to either an IC engine or compressor. The only difference between these two choices is the magnitude of the cylinder pressure used in the calculations. The default pressure for either device can also be changed in the box at lower right. The stroke cycle can be specified as 2 or 4 stroke. Two operating speeds must be specified, idle and redline. The midrange speed is the average of the two speeds supplied. One of three calculation speeds must be selected from the *Idle*, *Midrange*, and *Redline* options. The number of cylinders can be specified in the *No. Cylinders* box, and *Engine Type* can be selected from its pull-down menu. At this screen, only one cylinder of the engine is being calculated, so the choices of number of cylinders and engine type are not used until later when the engine is assembled. Choices made on this screen will be carried forward to the *Assemble* screen and also can be changed there if desired. The CG locations of crank and conrod, expressed as a percent of length are typed in boxes on the left. The masses of crank, conrod, and piston, must be supplied (in proper units) in the boxes at the right.

Text boxes in the *Engine Geometry Panel* on the left allow typing of the cylinder displacement, bore, stroke or crank radius, and L/R ratio. Cylinder displacement is the primary variable in this program. Any displacement volume can be achieved with an infinity of combinations of bore and stroke. To resolve this indeterminacy, bore is arbitrarily given precedence over stroke when the displacement is changed. That is, a change of displacement will force a change in stroke, leaving the bore unchanged. Changing the bore will force a change in stroke, keeping the specified displacement. Either the stroke or the crank radius may be changed, but either will force the other to change accordingly since stroke is always twice the crank radius, so only one need be input. The bore/stroke ratio is calculated and displayed at upper right.

The diameters of main pins and crankpins are used only to calculate an estimated friction torque in combination with an estimate of friction coefficient in the engine. The friction torque is calculated by multiplying the user-specified coefficient of friction by the forces calculated at the piston-cylinder interface and at the main pin and crankpin

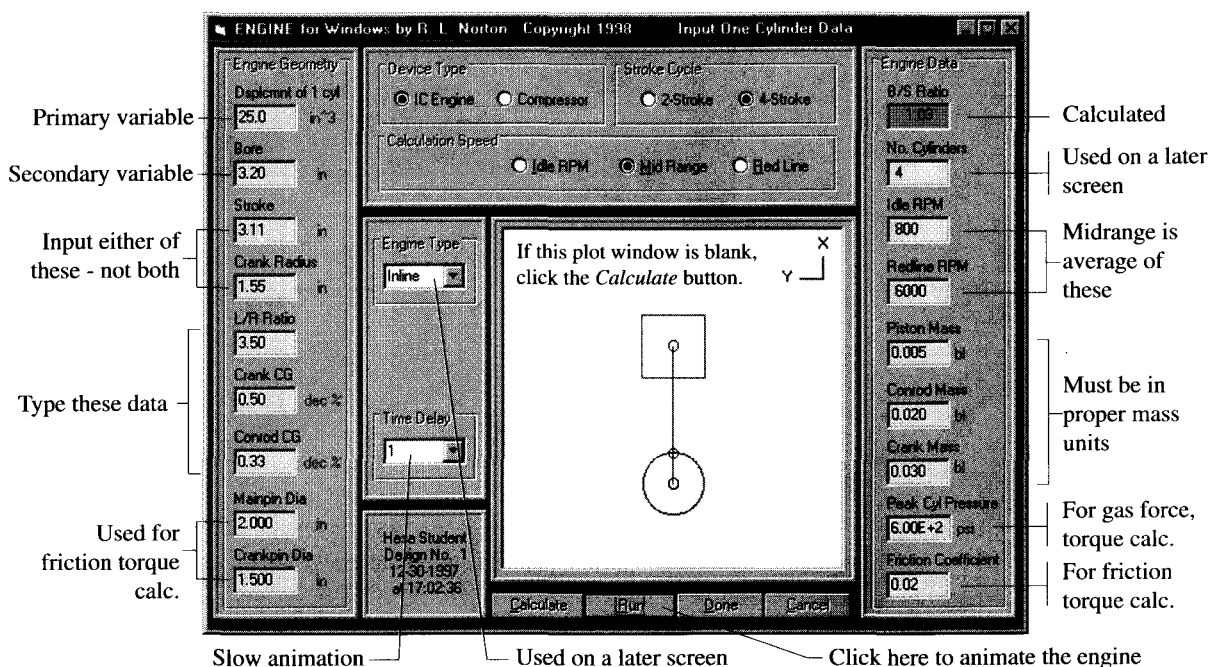


FIGURE A-20

Program ENGINE Input Screen for one-cylinder engine data

journals. These last two friction forces are multiplied by the journal radii supplied by the user to obtain friction torque. The piston friction creates friction torque through the geometry described in the gas torque equation 13.8 (p. 613). The other torque values (gas torque, inertia torque, and total torque) are **not** reduced in the program by the amount of the calculated friction torque, which is at best a crude estimate.

CALCULATION When all data are supplied, a click of the *Calculate* button will cause the piston position, velocity, and acceleration; the inertia forces and torques; and pin forces plus gas force and gas torque to be calculated for two revolutions of the crankshaft. At this screen, these data are computed only for one cylinder of the engine, regardless of how many cylinders were specified. The gas force and gas torque calculation is based on a built-in gas pressure curve similar to the one shown in Figure 13-4e (p. 602). This gas pressure function in the program is kept the same at all engine speeds as discussed in Section 13.1 (p. 600). Though this is not accurate in a thermodynamic sense, it is both necessary and appropriate for the purpose of comparing designs based solely on their kinematic and dynamic factors.

Calculations are done for all parameters at 3° increments over two crankshaft revolutions, giving 241 data points per variable. When calculations are complete, the plot window will show a schematic of the single-cylinder engine that can be animated with the *Run* button. The bore, stroke and conrod dimensions are to scale in the animation. The *Time Delay* value can be increased to slow the animation.

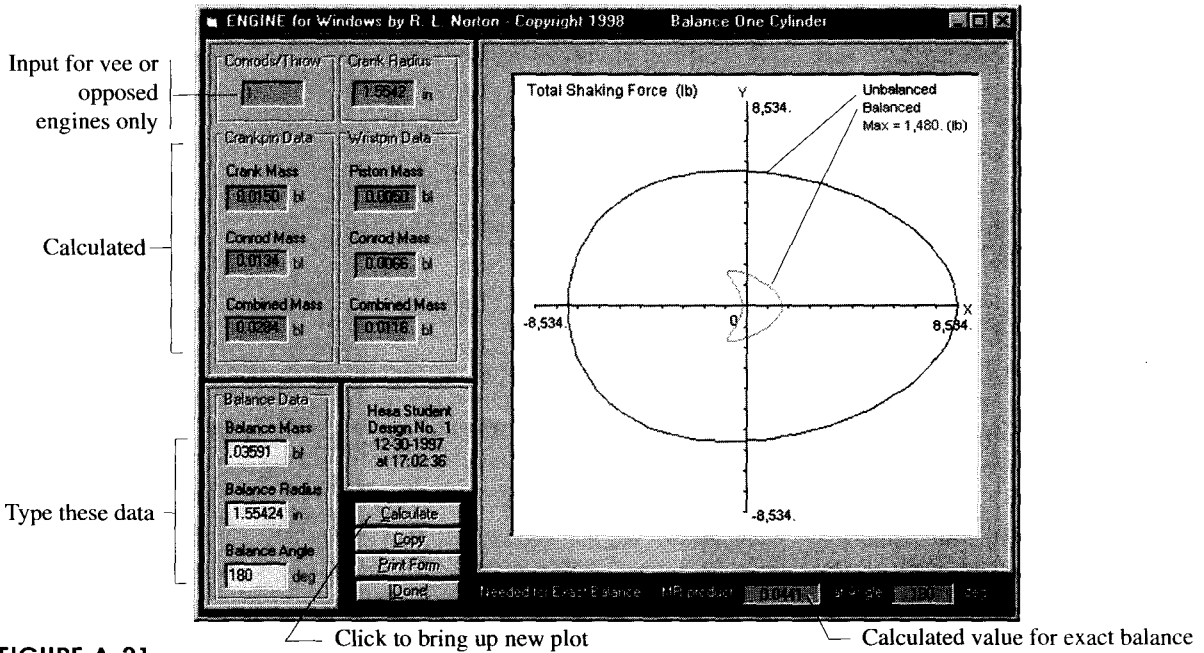


FIGURE A-21

Single-cylinder *Balance* screen from program ENGINE

The *Done* button returns you to the *Home* screen where you can use the *Plot* and *Print* buttons to display the results of the single-cylinder calculations. See Section A.2 (p. 727) for information on the *Print* and *Plot* screens.

Balancing the Crank (ENGINE *Balance* Screen)

The *Balance* button on the *Home* screen brings up the *Balance* screen shown in Figure A-21. The bottom of this screen displays the mass-radius product needed on the crank to cancel the primary component of unbalance due to the mass at the crankpin. The unbalanced shaking force is displayed as a hodograph in a plot window. Information on the amounts of mass estimated to be located at the crankpin and wrist pin is displayed in the sidebar. These data provide enough information to determine the counterweight parameters needed to either exactly balance or optimally overbalance the single-cylinder engine. Three text boxes at the bottom left of the screen allow input of the desired mass, radius, and angle of the counterweight proposed to be placed on the crankshaft.

Clicking the *Calculate* button recomputes the shaking force and torque with the added counterweight and superposes a hodograph of the new, balanced shaking force on the plot of the unbalanced force at the same scale so the improvement can be seen. See Chapter 13 for a discussion of the meaning and use of these data. Note that if the engine design has enough cylinders to allow a crankshaft arrangement that will cancel the inertial forces, then there may be no advantage to overbalancing the crank. But, for a single-cylinder engine and some two-cylinder engines (twins), overbalancing the crank can significantly reduce the shaking force.

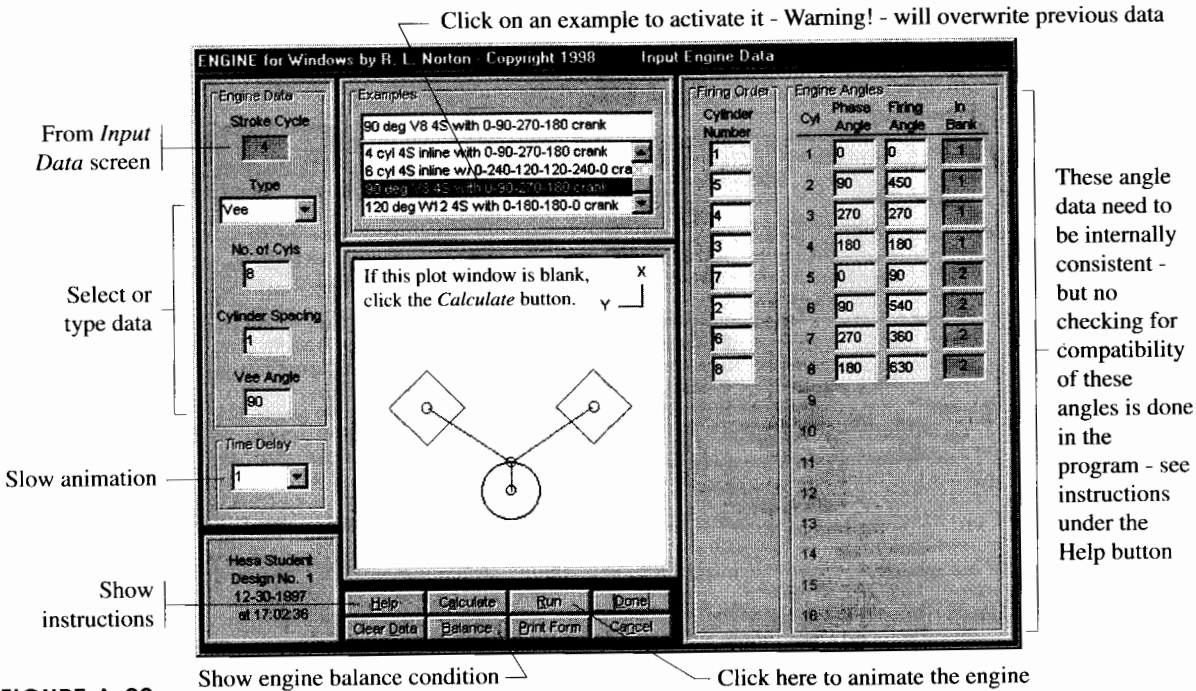


FIGURE A-22 The *Input Engine Data* screen in program *ENGINE* is used to assemble multicylinder engines

Assembling a Multicylinder Engine (*Engine Data* Screen)

Once the single-cylinder configuration is satisfactorily designed and balanced, the engine can be assembled. Figure A-22 shows the *Engine Data* screen used for assembly of a multicylinder engine. Its *Help* button brings up the information shown in Figure A-23, which points out that the crank phase diagram (called a *Power Chart* in program *ENGINE*) as defined in Section 14.2 (p. 644) and shown in Figures 14-8, 14-11, 14-13, 14-15, 14-17, and 14-23 (pp. 646-672) should be defined and drawn manually before proceeding with the assembly of the engine. The assumptions and conventions used in the program to number the cylinders and banks are also stated on these screens. These are the same conventions as defined in Section 14.7 (p. 661). The upper limit stated in item 7 of Figure A-23 for the acceptable range of power stroke angles will be either 360° for a two-stroke engine or 720° (as shown) for a four-stroke engine.

The firing order, the crankshaft phase angles, and the angles at which the cylinders fire (firing angles)-all angles in cylinder order-are typed in the boxes on the right of the *Engine Data* screen based on an arrangement according to the rules in Figure A-23. Note that the program does not do any internal check on the compatibility of these data. It will accept any combination of phase angles, firing orders, and power stroke angles you provide. It is up to you to ensure that these data are compatible and realistic.

The program also needs the distance between cylinders which is defaulted to one. This information is used to define the Z_i in equation 14.11 (p. 667). If you want to com-

+++++++ Warning ++++++

Before proceeding you need to work out, on paper, the crank's Phase Angles, the engine's Firing Order, and the Power Stroke (the angles at which the cylinders fire).

This program uses the Phase Angles and Firing Angles and Vee Angle to calculate the results. Firing Order is used as a pointer. Firing Order, it will make the results incorrect. It is your responsibility to ensure that all these factors agree!

This program requires that these rules be followed when designing and numbering your crankshaft phase and firing angles, and vee angle:

- 1 - The engine's front cylinder in the right bank is always number 1
- 2 - Number one cylinder is the reference cylinder for all others
- 3 - Number one cylinder always has a Crank Phase Angle of zero degrees
- 4 - Number one cylinder always fires first in Firing Order
- 5 - Thus the Power Stroke Angle for number one is also zero degrees
- 6 - Crank Phase Angles are always between 0 and 360 degrees
- 7 - Firing Angles are between 0 and 720 deg. and in ascending order
- 8 - Phase Angles shift to the right versus cylinder 1 on the Crank Phase Diagram
- 9 - The Vee Angle shifts the second bank to the Right on the Crank Phase Diagram
- 10 - Cylinders are numbered first down right bank and then down left bank

You must draw your crank phase diagram to decide these values before proceeding.

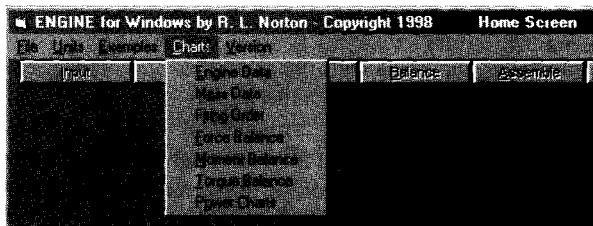
FIGURE A-23

Rules for assembling a multicylinder engine in program ENGINE

pute the correct magnitude of the shaking moment, the actual cylinder spacing of your design needs to be supplied in the box at the left of the screen in Figure A-22. If you only want to compare two or more engine designs on the basis of relative shaking moment, then the default value of one will suffice. The computation sums moments about the first cylinder, so its moment arm is always zero. The small offset between a vee engine's banks, due to having two conrods per crank throw, is ignored in the moment calculation.

Charts (Pull-Down Menu)

Figure A-24 shows the *Charts* pull-down menu on the *Home* screen. Several summaries of engine information are available there. The *Force Balance*, *Moment Balance*, and *Torque Balance* menu selections display the results of engine balance condition calculations, i.e., equations 14.3a and 14.3b (p. 648) for force balance; 14.7a and 14.7b (p. 650) for moment balance; and 14.5a, 14.5b, and 14.5c (p. 650) for torque balance. (These charts are also accessible from the *Engine Data* screen via its *Balance* button.)

**FIGURE A-24**

Charts available from pull-down menu in program ENGINE

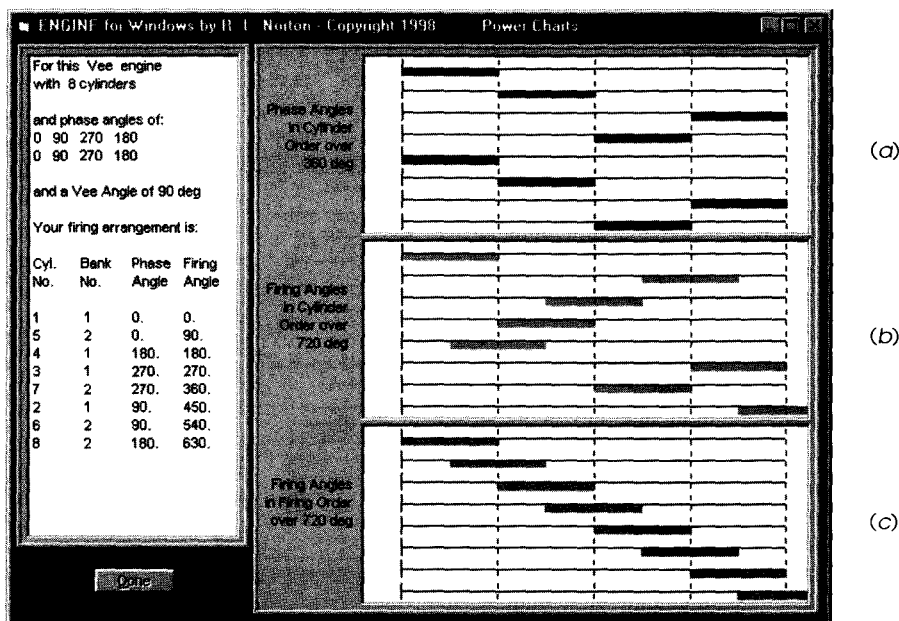


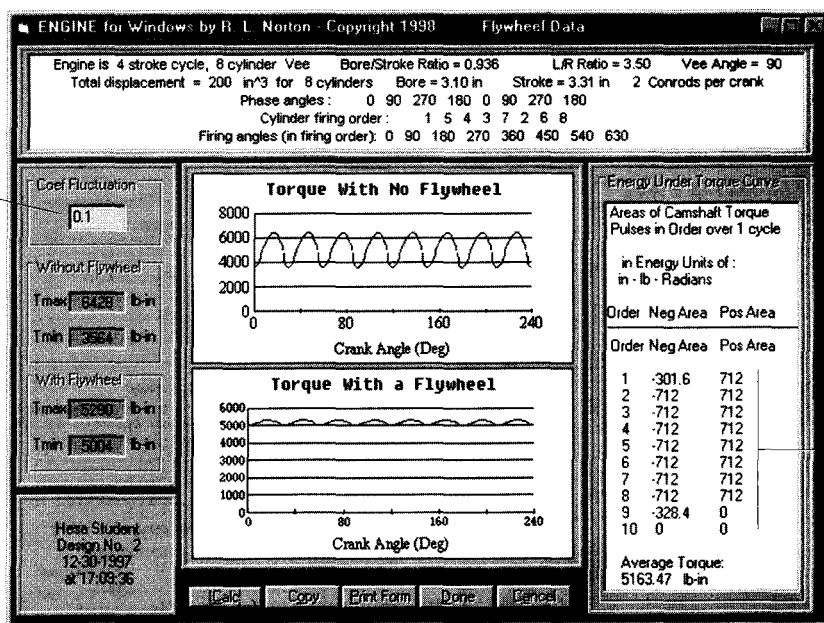
FIGURE A-25

Phase angle and power charts for a vee-eight engine from program ENGINE. Compare to Figure 14-23 (p. 672)

The **Power Chart** selection presents the three plots shown in Figure A-25. Figure A-25a is a plot of the crankshaft phase angles over 360° for the 90° vee eight engine designed in Section 14.7 (p. 661). Figure A-25b shows the power stroke angles in cylinder order over 720° for this vee eight. This is the same as the crank phase diagram shown in Figure 14-23 (p. 672) and indicates when the power pulses will occur in each cylinder during the cycle. Figure A-25c shows the power stroke angles in firing order over 720° . This third plot rearranges the order of the cylinders on the power chart to match that of the selected firing order. If these three factors, *crankshaft phase angles*, *firing order*, and *firing angles* have all been correctly chosen for compatibility, this third power chart plot will appear as a *staircase*. The power pulses should occur in equispaced steps across the cycle if even firing has been achieved. Thus these power chart plots can serve as a check on the correct choices of these factors. Note that the program only draws the boxes on the crank phase diagrams and power charts which represent the power strokes of each cylinder. The other piston strokes shown in Figures 14-8, 14-11, 14-13, 14-15, 14-17, and 14-23 (pp. 646-672) are not drawn.

Flywheel Calculations

Figure A-26 shows the *Flywheel Data* screen from program ENGINE. The only input to this screen is the flywheel coefficient of fluctuation which must be between 0 and 1. When the *Calc* button is clicked, it does the same pulse-by-pulse integration of the total torque curve as is done in program FOURBAR. See Section 11.11 and Figure 11-16 (p. 548). The areas under the pulses of the total torque curve are a measure of energy variation in the cycle. The flywheel size is based on these energy variations as defined in Sec-



0 < value < 1
to generate
flywheel
torque curve

Calculated
max and min
torques
before and
after flywheel

From a
piecewise
integration of
the torque
curve at top
left

Click here to see the plots

FIGURE A-26

Flywheel Data screen from program ENGINE for a vee-eight engine

tion 11.11. The total torque curve for this example vee-eight engine is shown in the upper plot of Figure A-26. The table of areas under its pulses is shown in the table at the right of the screen. Note that for a properly designed, even firing, multicylinder engine, the absolute values of the areas under each pulse will be equal. In that case the energy value needed for the flywheel calculation is simply the area under one pulse. These data can be used in equation 11.22 (p. 553) to compute the needed flywheel moment of inertia based on a chosen coefficient of fluctuation k . Program ENGINE then calculates a smoothed total torque function by multiplying the raw torque by the specified coefficient of fluctuation. The lower plot in Figure A-26 shows the flywheel smoothed torque for the example vee-eight engine with $k = 0.1$. The torque function with the flywheel added is nearly a constant value, which is desirable. The pulses due to the individual cylinder's explosions have been effectively masked by the flywheel. The maximum and minimum values of torque with and without the flywheel are shown in boxes at the left of the screen. Data on the engine design are displayed in the panel at the top of the screen.

A.9 PROGRAM MATRIX

Program *MATRIXfor Windows* solves a matrix equation of the form $A \times B = C$. A must be square and can have up to 16 rows and 16 columns. C must be a vector with as many elements as A has rows. When the data have all been typed in and checked, click the *Solve* button and it will compute the terms of vector B . The results can be printed to screen, printer, or a disk file in the same manner as described in Section A.2 (p. 727). If the matrix is singular or the solution fails for other reasons, an error message will be returned. No plotting or animation is provided in this program.